



PART
4
Tools and
Concepts

4 Probability and Probability
Distributions

CHAPTER 4

Probability and Probability Distributions

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4.1 How Probability Can Be Used in Making Inferences

We stated in Chapter 1 that a scientist uses inferential statistics to make statements about a population based on information contained in a sample of units selected from that population. Graphical and numerical descriptive techniques were presented in Chapter 3 as a means to summarize and describe a sample. However, a sample is not identical to the population from which it was selected. We need

to assess the degree of accuracy to which the sample mean, sample standard deviation, or sample proportion represent the corresponding population values.

Most management decisions must be made in the presence of uncertainty. Prices and designs for new automobiles must be selected on the basis of shaky forecasts of consumer preference, national economic trends, and competitive actions. The size and allocation of a hospital staff must be decided with limited information on patient load. The inventory of a product must be set in the face of uncertainty about demand. Probability is the language of uncertainty. Now let us examine probability, the mechanism for making inferences. This idea is probably best illustrated by an example.

Newsweek, in its June 20, 1998 issue, asks the question, “Who Needs Doctors? The Boom in Home Testing.” The article discusses the dramatic increase in medical screening tests for home use. The home-testing market has expanded beyond the two most frequently used tests, pregnancy and diabetes glucose monitoring, to a variety of diagnostic tests that were previously used only by doctors and certified laboratories. There is a DNA test to determine whether twins are fraternal or identical, a test to check cholesterol level, a screening test for colon cancer, and tests to determine whether your teenager is a drug user. However, the major question that needs to be addressed is how reliable are the testing kits? When a test indicates that a woman is not pregnant, what is the chance that the test is incorrect and the woman is truly pregnant? This type of incorrect result from a home test could translate into a woman not seeking the proper prenatal care in the early stages of her pregnancy.

Suppose a company states in its promotional materials that its pregnancy test provides correct results in 75% of its applications by pregnant women. We want to evaluate the claim, and so we select 20 women who have been determined by their physicians, using the best possible testing procedures, to be pregnant. The test is taken by each of the 20 women, and for all 20 women the test result is negative, indicating that none of the 20 is pregnant. What do you conclude about the company’s claim on the reliability of its test? Suppose you are further assured that each of the 20 women was in fact pregnant, as was determined several months after the test was taken.

If the company’s claim of 75% reliability was correct, we would have expected somewhere near 75% of the tests in the sample to be positive. However, none of the test results was positive. Thus, we would conclude that the company’s claim is probably false. Why did we fail to state with certainty that the company’s claim was false? Consider the possible setting. Suppose we have a large population consisting of millions of units, and 75% of the units are Ps for positives and 25% of the units are Ns for negatives. We randomly select 20 units from the population and count the number of units in the sample that are Ps. Is it possible to obtain a sample consisting of 0 Ps and 20 Ns? Yes, it is possible, *but it is highly improbable*. Later in this chapter we will compute the probability of such a sample occurrence.

To obtain a better view of the role that probability plays in making inferences from sample results to conclusions about populations, suppose the 20 tests result in 14 tests being positive—that is, a 70% correct response rate. Would you consider this result highly improbable and reject the company’s claim of a 75% correct response rate? How about 12 positives and 8 negatives, or 16 positives and 4 negatives? At what point do we decide that the result of the observed sample is so improbable, assuming the company’s claim is correct, that we disagree with its claim? To answer this question, we must know how to find the probability of

obtaining a particular sample outcome. Knowing this probability, we can then determine whether we agree or disagree with the company's claim. Probability is the tool that enables us to make an inference. Later in this chapter we will discuss in detail how the FDA and private companies determine the reliability of screening tests.

Because probability is the tool for making inferences, we need to define probability. In the preceding discussion, we used the term *probability* in its everyday sense. Let us examine this idea more closely.

Observations of phenomena can result in many different outcomes, some of which are more likely than others. Numerous attempts have been made to give a precise definition for the probability of an outcome. We will cite three of these.

classical interpretation

The first interpretation of probability, called the **classical interpretation of probability**, arose from games of chance. Typical probability statements of this type are, for example, “the probability that a flip of a balanced coin will show ‘heads’ is $1/2$ ” and “the probability of drawing an ace when a single card is drawn from a standard deck of 52 cards is $4/52$.” The numerical values for these probabilities arise from the nature of the games. A coin flip has two possible outcomes (a head or a tail); the probability of a head should then be $1/2$ (1 out of 2). Similarly, there are 4 aces in a standard deck of 52 cards, so the probability of drawing an ace in a single draw is $4/52$, or 4 out of 52.

outcome event

In the classical interpretation of probability, each possible distinct result is called an **outcome**; an **event** is identified as a collection of outcomes. The probability of an event E under the classical interpretation of probability is computed by taking the ratio of the number of outcomes, N_e , favorable to event E to the total number N of possible outcomes:

$$P(\text{event } E) = \frac{N_e}{N}$$

The applicability of this interpretation depends on the assumption that all outcomes are equally likely. If this assumption does not hold, the probabilities indicated by the classical interpretation of probability will be in error.

relative frequency interpretation

A second interpretation of probability is called the **relative frequency concept of probability**; this is an empirical approach to probability. If an experiment is repeated a large number of times and event E occurs 30% of the time, then .30 should be a very good approximation to the probability of event E . Symbolically, if an experiment is conducted n different times and if event E occurs on n_e of these trials, then the probability of event E is approximately

$$P(\text{event } E) \approx \frac{n_e}{n}$$

We say “approximate” because we think of the actual probability $P(\text{event } E)$ as the relative frequency of the occurrence of event E over a very large number of observations or repetitions of the phenomenon. The fact that we can check probabilities that have a relative frequency interpretation (by simulating many repetitions of the experiment) makes this interpretation very appealing and practical.

The third interpretation of probability can be used for problems in which it is difficult to imagine a repetition of an experiment. These are “one-shot” situa-

subjective interpretation

tions. For example, the director of a state welfare agency who estimates the probability that a proposed revision in eligibility rules will be passed by the state legislature would not be thinking in terms of a long series of trials. Rather, the director would use a **personal** or **subjective probability** to make a one-shot statement of belief regarding the likelihood of passage of the proposed legislative revision. The problem with subjective probabilities is that they can vary from person to person and they cannot be checked.

Of the three interpretations presented, the relative frequency concept seems to be the most reasonable one because it provides a practical interpretation of the probability for most events of interest. Even though we will never run the necessary repetitions of the experiment to determine the exact probability of an event, the fact that we can check the probability of an event gives meaning to the relative frequency concept. Throughout the remainder of this text we will lean heavily on this interpretation of probability.

EXERCISES

Applications

- 4.1** Indicate which interpretation of the probability statement seems most appropriate.
- The National Angus Association has stated that there is a 60/40 chance that wholesale beef prices will rise by the summer—that is, a .60 probability of an increase and a .40 probability of a decrease.
 - The quality control section of a large chemical manufacturing company has undertaken an intensive process-validation study. From this study, the QC section claims that the probability that the shelf life of a newly released batch of chemical will exceed the minimal time specified is .998.
 - A new blend of coffee is being contemplated for release by the marketing division of a large corporation. Preliminary marketing survey results indicate that 550 of a random sample of 1,000 potential users rated this new blend better than a brand-name competitor. The probability of this happening is approximately .001, assuming that there is actually no difference in consumer preference for the two brands.
 - The probability that a customer will receive a package the day after it was sent by a business using an “overnight” delivery service is .92.
 - The sportscaster in College Station, Texas states that the probability that the Aggies will win their football game against the University of Florida is .75.
 - The probability of a nuclear power plant having a meltdown on a given day is .00001.
 - If a customer purchases a single ticket for the Texas lottery, the probability of that ticket being the winning ticket is $1/15,890,700$.
- 4.2** Give your own personal probability for each of the following situations. It would be instructive to tabulate these probabilities for the entire class. In which cases did you have large disagreements?
- The federal income tax will be eliminated.
 - You will receive an A in this course.
 - Two or more individuals in the classroom have the same birthday.
 - An asteroid will strike the planet earth in the next year.
 - A woman will be elected as vice president or president of the United States in the next presidential election.

4.2

Finding the Probability of an Event

In the preceding section, we discussed three different interpretations of probability. In this section, we will use the classical interpretation and the relative frequency

concept to illustrate the computation of the probability of an outcome or event. Consider an experiment that consists of tossing two coins, a penny and then a dime, and observing the upturned faces. There are four possible outcomes:

TT: tails for both coins

TH: a tail for the penny, a head for the dime

HT: a head for the penny, a tail for the dime

HH: heads for both coins

What is the probability of observing the event exactly one head from the two coins?

This probability can be obtained easily if we can assume that all four outcomes are equally likely. In this case, that seems quite reasonable. There are $N = 4$ possible outcomes, and $N_e = 2$ of these are favorable for the event of interest, observing exactly one head. Hence, by the classical interpretation of probability,

$$P(\text{exactly 1 head}) = \frac{2}{4} = \frac{1}{2}$$

Because the event of interest has a relative frequency interpretation, we could also obtain this same result empirically, using the relative frequency concept. To demonstrate how relative frequency can be used to obtain the probability of an event, we will use the ideas of simulation. Simulation is a technique that produces outcomes having the same probability of occurrence as the real situation events. The computer is a convenient tool for generating these outcomes. Suppose we wanted to simulate 1,000 tosses of the two coins. We can use a computer program such as *SAS* or *Minitab* to simulate the tossing of a pair of coins. The program has a random number generator. We will designate an even number as *H* and an odd number as *T*. Since there are five even and five odd single-digit numbers, the probability of obtaining an even number is $5/10 = .5$, which is the same as the probability of obtaining an odd number. Thus, we can request 500 pairs of single-digit numbers. This set of 500 pairs of numbers will represent 500 tosses of the two coins, with the first digit representing the outcome of tossing the penny and the second digit representing the outcome of tossing the dime. For example, the pair (3, 6) would represent a tail for the penny and a head for the dime. Using version 13 of Minitab, the following steps will generate 1,000 randomly selected numbers from 0 to 9:

1. Select **Calc** from the toolbar
2. Select **Random Data** from list
3. Select **Integer** from list
4. Generate **20** rows of data
5. Store in column(s): **c1-c50**
6. Minimum value: **0**
7. Maximum value: **9**

The preceding steps will produce 1,000 random single-digit numbers that can then be paired to yield 500 pairs of single-digit numbers. (Most computer packages contain a random number generator that can be used to produce similar results.)

25 32 70 15 96 87 80 43 15 77 89 51 08 36 29 55 42 86 45 93 68 72 49 99 37
82 81 58 50 85 27 99 41 10 31 42 35 50 02 68 33 50 93 73 62 15 15 90 97 24
46 86 89 82 20 23 63 59 50 40 32 72 59 62 58 53 01 85 49 27 31 48 53 07 78
15 81 39 83 79 21 88 57 35 33 49 37 85 42 28 38 50 43 82 47 01 55 42 02 52
66 44 15 40 29 73 11 06 79 81 49 64 32 06 07 31 07 78 73 07 26 36 39 20 14
48 20 27 73 53 21 44 16 00 33 43 95 21 08 19 60 68 30 99 27 22 74 65 22 05
26 79 54 64 94 01 21 47 86 94 24 41 06 81 16 07 30 34 99 54 68 37 38 71 79
86 12 83 09 27 60 49 54 21 92 64 57 07 39 04 66 73 76 74 93 50 56 23 41 23
18 87 21 48 75 63 09 97 96 86 85 68 65 35 92 40 57 87 82 71 04 16 01 03 45
52 79 14 12 94 51 39 40 42 17 32 94 42 34 68 17 39 32 38 03 75 56 79 79 57
07 40 96 46 22 04 12 90 80 71 46 11 18 81 54 95 47 72 06 07 66 05 59 34 81
66 79 83 82 62 20 75 71 73 79 48 86 83 74 04 13 36 87 96 11 39 81 59 41 70
21 47 34 02 05 73 71 57 64 58 05 16 57 27 66 92 97 68 18 52 09 45 34 80 57
87 22 18 65 66 18 84 31 09 38 05 67 10 45 03 48 52 48 33 36 00 49 39 55 35
70 84 50 37 58 41 08 62 42 64 02 29 33 68 87 58 52 39 98 78 72 13 13 15 96
57 32 98 05 83 39 13 39 37 08 17 01 35 13 98 66 89 40 29 47 37 65 86 73 42
85 65 78 05 24 65 24 92 03 46 67 48 90 60 02 61 21 12 80 70 35 15 40 52 76
29 11 45 22 38 33 32 52 17 20 03 26 34 18 85 46 52 66 63 30 84 53 76 47 21
42 97 56 38 41 87 14 43 30 35 99 06 76 67 00 47 83 32 52 42 48 51 69 15 18
08 30 37 89 17 89 23 58 13 93 17 44 09 08 61 05 35 44 91 89 35 15 06 39 27

The summary of the simulation of the 500 tosses is shown in Table 4.1.

TABLE 4.1
Simulation of tossing a
penny and a dime 500 times

Event	Outcome of Simulation	Frequency	Relative Frequency
TT	(Odd, Odd)	129	$129/500 = .258$
TH	(Odd, Even)	117	$117/500 = .234$
HT	(Even, Odd)	125	$125/500 = .250$
HH	(Even, Even)	129	$129/500 = .258$

Note that this approach yields simulated probabilities that are nearly in agreement with our intuition; that is, intuitively we might expect these outcomes to be equally likely. Thus, each of the four outcomes should occur with a probability equal to $1/4$, or $.25$. This assumption was made for the classical interpretation. We will show in Chapter 10 that in order to be 95% certain that the simulated probabilities are within $.01$ of the true probabilities, the number of tosses should be at least 7,500 and not 500 as we used previously.

If we wish to find the probability of tossing two coins and observing exactly one head, we have, from Table 4.1,

$$P(\text{exactly 1 head}) \approx \frac{117 + 125}{500} = .484$$

This is very close to the theoretical probability, which we have shown to be $.5$.

Note that we could easily modify our example to accommodate the tossing of an unfair coin. Suppose we are tossing a penny that is weighted so that the probability of a head occurring in a toss is $.70$ and the probability of a tail is $.30$. We could designate an H outcome whenever one of the random digits 0, 1, 2, 3,

4, 5, or 6 occurs and a T outcome whenever one of the digits 7, 8, or 9 occurs. The same simulation program can be run as before, but we would interpret the output differently.

EXERCISES**Applications****Edu.**

4.3 Suppose an exam consists of 20 true-or-false questions. A student takes the exam by guessing the answer to each question. What is the probability that the student correctly answers 15 or more of the questions? [*Hint:* Use a simulation approach. Generate a large number (2,000 or more sets) of 20 single-digit numbers. Each number represents the answer to one of the questions on the exam, with even digits representing correct answers and odd digits representing wrong answers. Determine the relative frequency of the sets having 15 or more correct answers.]

Med.

4.4 The example in Section 4.1 considered the reliability of a screening test. Suppose we wanted to simulate the probability of observing at least 15 positive results and 5 negative results in a set of 20 results, when the probability of a positive result was claimed to be .75. Use a random number generator to simulate the running of 20 screening tests.

- a. Let a two-digit number represent an individual running of the screening test. Which numbers represent a positive outcome of the screening test? Which numbers represent a negative outcome?
- b. If we generate 2,000 sets of 20 two-digit numbers, how can the outcomes of this simulation be used to approximate the probability of obtaining at least 15 positive results in the 20 runnings of the screening test?

4.3**Basic Event Relations and Probability Laws**

The probability of an event, say event A , will always satisfy the property

$$0 \leq P(A) \leq 1$$

that is, the probability of an event lies anywhere in the interval from 0 (the occurrence of the event is impossible) to 1 (the occurrence of an event is a “sure thing”).

either A or B occurs

Suppose A and B represent two experimental events and you are interested in a new event, the event that **either A or B occurs**. For example, suppose that we toss a pair of dice and define the following events:

A : A total of 7 shows

B : A total of 11 shows

Then the event “either A or B occurs” is the event that you toss a total of either 7 or 11 with the pair of dice.

mutually exclusive

Note that, for this example, the events A and B are **mutually exclusive**; that is, if you observe event A (a total of 7), you could not at the same time observe event B (a total of 11). Thus, if A occurs, B cannot occur (and vice versa).

DEFINITION 4.1

Two events A and B are said to be **mutually exclusive** if (when the experiment is performed a single time) the occurrence of one of the events excludes the possibility of the occurrence of the other event.

The concept of mutually exclusive events is used to specify a second property that the probabilities of events must satisfy. When two events are mutually exclusive, then the probability that either one of the events will occur is the sum of the event probabilities.

DEFINITION 4.2

If two events, A and B , are mutually exclusive, the **probability** that either event occurs is $P(\text{either } A \text{ or } B) = P(A) + P(B)$.

Definition 4.2 is a special case of the union of two events, which we will soon define.

The definition of additivity of probabilities for mutually exclusive events can be extended beyond two events. For example, when we toss a pair of dice, the sum S of the numbers appearing on the dice can assume any one of the values $S = 2, 3, 4, \dots, 11, 12$. On a single toss of the dice, we can observe only one of these values. Therefore, the values 2, 3, $\dots$, 12 represent mutually exclusive events. If we want to find the probability of tossing a sum less than or equal to 4, this probability is

$$P(S \leq 4) = P(2) + P(3) + P(4)$$

For this particular experiment, the dice can fall in 36 different equally likely ways. We can observe a 1 on die 1 and a 1 on die 2, denoted by the symbol $(1, 1)$. We can observe a 1 on die 1 and a 2 on die 2, denoted by $(1, 2)$. In other words, for this experiment, the possible outcomes are

$(1, 1)$	$(2, 1)$	$(3, 1)$	$(4, 1)$	$(5, 1)$	$(6, 1)$
$(1, 2)$	$(2, 2)$	$(3, 2)$	$(4, 2)$	$(5, 2)$	$(6, 2)$
$(1, 3)$	$(2, 3)$	$(3, 3)$	$(4, 3)$	$(5, 3)$	$(6, 3)$
$(1, 4)$	$(2, 4)$	$(3, 4)$	$(4, 4)$	$(5, 4)$	$(6, 4)$
$(1, 5)$	$(2, 5)$	$(3, 5)$	$(4, 5)$	$(5, 5)$	$(6, 5)$
$(1, 6)$	$(2, 6)$	$(3, 6)$	$(4, 6)$	$(5, 6)$	$(6, 6)$

As you can see, only one of these events, $(1, 1)$, will result in a sum equal to 2. Therefore, we would expect a 2 to occur with a relative frequency of $1/36$ in a long series of repetitions of the experiment, and we let $P(2) = 1/36$. The sum $S = 3$ will occur if we observe either of the outcomes $(1, 2)$ or $(2, 1)$. Therefore, $P(3) = 2/36 = 1/18$. Similarly, we find $P(4) = 3/36 = 1/12$. It follows that

$$P(S \leq 4) = P(2) + P(3) + P(4) = \frac{1}{36} + \frac{1}{18} + \frac{1}{12} = \frac{1}{6}$$

complement

A third property of event probabilities concerns an event and its **complement**.

DEFINITION 4.3

The **complement** of an event A is the event that A *does not* occur. The complement of A is denoted by the symbol $\bar{A}$.

Thus, if we define the complement of an event A as a new event—namely, “ A does not occur”—it follows that

$$P(A) + P(\bar{A}) = 1$$

For an example, refer again to the two-coin-toss experiment. If, in many repetitions of the experiment, the proportion of times you observe event A , “two heads show,” is $1/4$, then it follows that the proportion of times you observe the event $\bar{A}$, “two heads do not show,” is $3/4$. Thus, $P(A)$ and $P(\bar{A})$ will always sum to 1.

We can summarize the three properties that the probabilities of events must satisfy as follows:

Properties of Probabilities

If A and B are any two mutually exclusive events associated with an experiment, then $P(A)$ and $P(B)$ must satisfy the following properties:

1. $0 \leq P(A) \leq 1$ and $0 \leq P(B) \leq 1$
2. $P(\text{either } A \text{ or } B) = P(A) + P(B)$
3. $P(A) + P(\bar{A}) = 1$ and $P(B) + P(\bar{B}) = 1$

union intersection

We can now define two additional event relations: the **union** and the **intersection** of two events.

DEFINITION 4.4

The **union** of two events A and B is the set of all outcomes that are included in either A or B (or both). The union is denoted as $A \cup B$.

DEFINITION 4.5

The **intersection** of two events A and B is the set of all outcomes that are included in both A and B . The intersection is denoted as $A \cap B$.

These definitions along with the definition of the complement of an event formalize some simple concepts. The event $\bar{A}$ occurs when A *does not*; $A \cup B$ occurs when either A or B occurs; $A \cap B$ occurs when A *and* B occur.

The additivity of probabilities for mutually exclusive events, called the *addition law for mutually exclusive events*, can be extended to give the general addition law.

DEFINITION 4.6

Consider two events A and B ; the **probability of the union** of A and B is

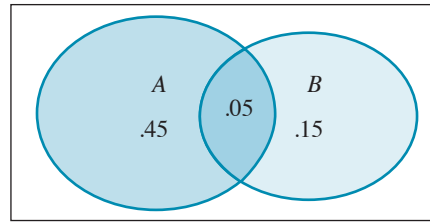
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

EXAMPLE 4.1

Events and event probabilities are shown in the Venn diagram in Figure 4.1. Use this diagram to determine the following probabilities:

- a. $P(A)$, $P(\bar{A})$
- b. $P(B)$, $P(\bar{B})$
- c. $P(A \cap B)$
- d. $P(A \cup B)$

FIGURE 4.1



Solution From the Venn diagram, we are able to determine the following probabilities:

- a. $P(A) = .5$, therefore $P(\bar{A}) = 1 - .5 = .5$
- b. $P(B) = .2$, therefore $P(\bar{B}) = 1 - .2 = .8$
- c. $P(A \cap B) = .05$
- d. $P(A \cup B) = P(A) + P(B) - P(A \cap B) = .5 + .2 - .05 = .65$

4.4

Conditional Probability and Independence

Consider the following situation: The examination of a large number of insurance claims, categorized according to type of insurance and whether the claim was fraudulent, produced the results shown in Table 4.2. Suppose you are responsible for checking insurance claims—in particular, for detecting fraudulent claims—and you examine the next claim that is processed. What is the probability of the event F , “the claim is fraudulent”? To answer the question, you examine Table 4.2 and note that 10% of all claims are fraudulent. Thus, assuming that the percentages given in the table are reasonable approximations to the true probabilities of receiving specific types of claims, it follows that $P(F) = .10$. Would you say that the risk that you face a fraudulent claim has probability .10? We think not, because you have additional information that may affect the assessment of $P(F)$. This additional information concerns the type of policy you were examining (fire, auto, or other).

Suppose that you have the additional information that the claim was associated with a fire policy. Checking Table 4.2, we see that 20% (or .20) of all claims are associated with a fire policy and that 6% (or .06) of all claims are fraudulent fire policy claims. Therefore, it follows that the probability that the claim is

TABLE 4.2
Categorization of insurance
claims

Category	Type of Policy (%)			Total %
	Fire	Auto	Other	
Fraudulent	6	1	3	10
Nonfraudulent	14	29	47	90
Total	20	30	50	100

fraudulent, given that you know the policy is a fire policy, is

$$\begin{aligned} P(F|\text{fire policy}) &= \frac{\text{proportion of claims that are fraudulent fire policy claims}}{\text{proportion of claims that are against fire policies}} \\ &= \frac{.06}{.20} = .30 \end{aligned}$$

conditional probability

This probability, $P(F|\text{fire policy})$, is called a **conditional probability** of the event F —that is, the probability of event F given the fact that the event “fire policy” has already occurred. This tells you that 30% of all fire policy claims are fraudulent. The vertical bar in the expression $P(F|\text{fire policy})$ represents the phrase “given that,” or simply “given.” Thus, the expression is read, “the probability of the event F given the event fire policy.”

unconditional probability

The probability $P(F) = .10$, called the **unconditional** or **marginal probability** of the event F , gives the proportion of times a claim is fraudulent—that is, the proportion of times event F occurs in a very large (infinitely large) number of repetitions of the experiment (receiving an insurance claim and determining whether the claim is fraudulent). In contrast, the conditional probability of F , given that the claim is for a fire policy, $P(F|\text{fire policy})$, gives the proportion of fire policy claims that are fraudulent. Clearly, the conditional probabilities of F , given the types of policies, will be of much greater assistance in measuring the risk of fraud than the unconditional probability of F .

DEFINITION 4.7

Consider two events A and B with nonzero probabilities, $P(A)$ and $P(B)$. The **conditional probability** of event A given event B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

The conditional probability of event B given event A is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

This definition for conditional probabilities gives rise to what is referred to as the *multiplication law*.

DEFINITION 4.8

The **probability of the intersection** of two events A and B is

$$\begin{aligned} P(A \cap B) &= P(A)P(B|A) \\ &= P(B)P(A|B) \end{aligned}$$

The only difference between Definitions 4.7 and 4.8, both of which involve conditional probabilities, relates to what probabilities are known and what needs to be calculated. When the intersection probability $P(A \cap B)$ and the individual probability $P(A)$ are known, we can compute $P(B|A)$. When we know $P(A)$ and $P(B|A)$, we can compute $P(A \cap B)$.

EXAMPLE 4.2

Two supervisors are to be selected as safety representatives within the company. Given that there are six supervisors in research and four in development, and each group of two supervisors has the same chance of being selected, find the probability of choosing both supervisors from research.

Solution Let A be the event that the first supervisor selected is from research, and let B be the event that the second supervisor is also from research. Clearly, we want $P(A \text{ and } B) = P(A \cap B) = P(B|A)P(A)$.

For this example,

$$P(A) = \frac{\text{number of research supervisors}}{\text{number of supervisors}} = \frac{6}{10}$$

and

$$P(B|A) = \frac{\text{number of research supervisors after one research supervisor was selected}}{\text{number of supervisors after one supervisor was selected}} = \frac{5}{9}$$

Thus,

$$P(A \cap B) = P(A)P(B|A) = \frac{6}{10} \left(\frac{5}{9} \right) = \frac{30}{90} = .333$$

Thus the probability of choosing both supervisors from research is .333, assuming that each group of two has the same chance of being selected.

Suppose that the probability of event A is the same whether event B has or has not occurred; that is, suppose

$$P(A|B) = P(A)$$

independent events

Then we say that the occurrence of event A is not dependent on the occurrence of event B or, simply, that A and B are **independent events**. When $P(A|B) \neq P(A)$, the occurrence of A depends on the occurrence of B , and events A and B are said to be **dependent events**.

dependent events

DEFINITION 4.9

Two events A and B are **independent events** if

$$P(A|B) = P(A) \quad \text{or} \quad P(B|A) = P(B)$$

(Note: You can show that if $P(A|B) = P(A)$, then $P(B|A) = P(B)$, and vice versa.)

Definition 4.9 leads to a special case of $P(A \cap B)$. When events A and B are independent, it follows that

$$P(A \cap B) = P(A)P(B|A) = P(A)P(B)$$

The concept of independence is of particular importance in sampling. Later in the text, we will discuss drawing samples from two (or more) populations to

independent samples

compare the population means, variances, or some other population parameters. For most of these applications, we will select samples in such a way that the observed values in one sample are independent of the values that appear in another sample. We call these **independent samples**.

EXERCISES

Basic Techniques

- 4.5** A coin is to be flipped three times. List the possible outcomes in the form (result on toss 1, result on toss 2, result on toss 3).
- 4.6** In Exercise 4.5, assume that each one of the outcomes has probability $1/8$ of occurring. Find the probability of
- A : observing exactly 1 head
 - B : observing 1 or more heads
 - C : observing no heads
- 4.7** For Exercise 4.6:
- Compute the probability of the complement of event A , event B , and event C .
 - Determine whether events A and B are mutually exclusive.
- 4.8** Determine the following conditional probabilities for the events of Exercise 4.6.
- $P(A|B)$
 - $P(A|C)$
 - $P(B|C)$
- 4.9** Refer to Exercise 4.8. Are events A and B independent? Why or why not? What about A and C ? What about B and C ?
- 4.10** A die is to be rolled and we are to observe the number that falls face up. Find the probabilities for these events:
- A : observe a 6
 - B : observe an even number
 - C : observe a number greater than 2
 - D : observe an even number and a number greater than 2
- 4.11** Refer to Exercise 4.10. Which of the events (A , B , and C) are independent? Which are mutually exclusive?
- 4.12** Consider the following outcomes for an experiment:

Outcome	1	2	3	4	5
Probability	.20	.25	.15	.10	.30

Let event A consist of outcomes 1, 3, and 5 and event B consist of outcomes 4 and 5.

- Find $P(A)$ and $P(B)$.
 - Find $P(\text{both } A \text{ and } B \text{ occur})$.
 - Find $P(\text{either } A \text{ or } B \text{ occurs})$.
- 4.13** Refer to Exercise 4.12. Does $P(\text{either } A \text{ or } B \text{ occurs}) = P(A) + P(B)$? Why or why not?

Applications

Edu.

4.14 A student has to have an accounting course and an economics course the next term. Assuming there are no schedule conflicts, describe the possible outcomes for selecting one section of the accounting course and one of the economics course if there are four possible accounting sections and three possible economics sections.

Engin.

4.15 The emergency room of a hospital has two backup generators, either of which can supply enough electricity for basic hospital operations. We define events A and B as follows:

event A : generator 1 works properly
 event B : generator 2 works properly

Describe the following events in words:

- complement of A
- $B|A$
- either A or B

- H.R. 4.16** A survey of a number of large corporations gave the following probability table for events related to the offering of a promotion involving a transfer.

Promotion/ Transfer	Married		Unmarried	Total
	Two-Career Marriage	One-Career Marriage		
Rejected	.184	.0555	.0170	.2565
Accepted	.276	.3145	.1530	.7435
Total	.46	.37	.17	

Use the probabilities to answer the following questions:

- What is the probability that a professional (selected at random) would accept the promotion? Reject it?
 - What is the probability that a professional (selected at random) is part of a two-career marriage? A one-career marriage?
- Bus. 4.17** An institutional investor is considering a large investment in two of five companies. Suppose that, unknown to the investor, two of the five firms are on shaky ground with regard to the development of new products.
- List the possible outcomes for this situation.
 - Determine the probability of choosing two of the three firms that are on better ground.
 - What is the probability of choosing one of two firms on shaky ground?
 - What is the probability of choosing the two shakiest firms?
- Soc. 4.18** A survey of workers in two manufacturing sites of a firm included the following question: How effective is management in responding to legitimate grievances of workers? The results are shown here.

	Number Surveyed	Number Responding “Poor”
Site 1	192	48
Site 2	248	80

Let A be the event the worker comes from Site 1 and B be the event the response is “poor.” Compute $P(A)$, $P(B)$, and $P(A \cap B)$.

- 4.19** Refer to Exercise 4.16.
- Are events A and B independent?
 - Find $P(B|A)$ and $P(B|\bar{A})$. Are they equal?
- H.R. 4.20** A large corporation has spent considerable time developing employee performance rating scales to evaluate an employee’s job performance on a regular basis, so major adjustments can be made when needed and employees who should be considered for a “fast track” can be isolated. Keys to this latter determination are ratings on the ability of an employee to perform to his or her capabilities and on his or her formal training for the job.

Workload Capacity	Formal Training			
	None	Little	Some	Extensive
Low	.01	.02	.02	.04
Medium	.05	.06	.07	.10
High	.10	.15	.16	.22

The probabilities for being placed on a fast track are as indicated for the 12 categories of workload capacity and formal training. The following three events (A , B , and C) are defined:

- A : an employee works at the high-capacity level
- B : an employee falls into the highest (extensive) formal training category
- C : an employee has little or no formal training and works below high capacity

- a. Find $P(A)$, $P(B)$, and $P(C)$.
- b. Find $P(A|B)$, $P(A|\bar{B})$, and $P(\bar{B}|C)$.
- c. Find $P(A \cup B)$, $P(A \cap C)$, and $P(B \cap C)$.

Bus. **4.21** The utility company in a large metropolitan area finds that 70% of its customers pay a given monthly bill in full.

- a. Suppose two customers are chosen at random from the list of all customers. What is the probability that both customers will pay their monthly bill in full?
- b. What is the probability that at least one of them will pay in full?

4.22 Refer to Exercise 4.21. A more detailed examination of the company records indicates that 95% of the customers who pay one monthly bill in full will also pay the next monthly bill in full; only 10% of those who pay less than the full amount one month will pay in full the next month.

- a. Find the probability that a customer selected at random will pay two consecutive months in full.
- b. Find the probability that a customer selected at random will pay neither of two consecutive months in full.
- c. Find the probability that a customer chosen at random will pay exactly one month in full.

4.5 Bayes' Formula

false positive
false negative

In this section, we will show how Bayes' Formula can be used to update conditional probabilities by using sample data when available. These "updated" conditional probabilities are useful in decision making. A particular application of these techniques involves the evaluation of diagnostic tests. Suppose a meat inspector must decide whether a randomly selected meat sample contains *E. coli* bacteria. The inspector conducts a diagnostic test. Ideally, a positive result (Pos) would mean that the meat sample actually has *E. coli*, and a negative result (Neg) would imply that the meat sample is free of *E. coli*. However, the diagnostic test is occasionally in error. The results of the test may be a **false positive**, for which the test's indication of *E. coli* presence is incorrect, or a **false negative**, for which the test's conclusion of *E. coli* absence is incorrect. Large-scale screening tests are conducted to evaluate the accuracy of a given diagnostic test. For example, *E. coli* (E) is placed in 10,000 meat samples, and the diagnostic test yields a positive result for 9,500 samples and a negative result for 500 samples; that is, there are 500 false negatives out of the 10,000 tests. Another 10,000 samples have all traces of *E. coli* (NE) removed, and the diagnostic test yields a positive result for 100 samples and a negative result for 9,900 samples. There are 100 false positives out of the 10,000 tests. We can summarize the results in the following table:

Diagnostic Test Result	Meat Sample Status	
	E	NE
Positive	9,500	100
Negative	500	9,900
Total	10,000	10,000

Evaluation of test results is as follows:

$$\text{True positive rate} = P(\text{Pos} | E) = \frac{9,500}{10,000} = .95$$

$$\text{False positive rate} = P(\text{Pos} | NE) = \frac{100}{10,000} = .01$$

$$\text{True negative rate} = P(\text{Neg} | NE) = \frac{9,900}{10,000} = .99$$

$$\text{False negative rate} = P(\text{Neg} | E) = \frac{500}{10,000} = .05$$

sensitivity
specificity

The **sensitivity** of the diagnostic test is the true positive rate, and the **specificity** of the diagnostic test is the true negative rate.

The primary question facing the inspector is to evaluate the probability of *E. coli* being present in the meat sample when the test yields a positive result; that is, the inspector needs to know $P(E | \text{Pos})$. Bayes' Formula answers this question, as the following calculations show. To make this calculation, we need to know the *rate* of *E. coli* in the type of meat being inspected. For this example, suppose that *E. coli* is present in 4.5% of all meat samples; that is, *E. coli* has prevalence $P(E) = .045$. We can then compute $P(E | \text{Pos})$ as follows:

$$\begin{aligned} P(E | \text{Pos}) &= \frac{P(E \cap \text{Pos})}{P(\text{Pos})} = \frac{P(E \cap \text{Pos})}{P(E \cap \text{Pos}) + P(NE \cap \text{Pos})} \\ &= \frac{P(\text{Pos} | E)P(E)}{P(\text{Pos} | E)P(E) + P(\text{Pos} | NE)P(NE)} \\ &= \frac{(.95)(.045)}{(.95)(.045) + (.01)(1 - .045)} = .817 \end{aligned}$$

Thus, *E. coli* is truly present in 81.7% of the tested samples in which a positive test result occurs. Also, we can conclude that 18.3% of the tested samples indicated *E. coli* was present when in fact there was no *E. coli* in the meat sample.

EXAMPLE 4.3

A book club classifies members as heavy, medium, or light purchasers, and separate mailings are prepared for each of these groups. Overall, 20% of the members are heavy purchasers, 30% medium, and 50% light. A member is not classified into a group until 18 months after joining the club, but a test is made of the feasibility of using the first 3 months' purchases to classify members. The following percentages are obtained from existing records of individuals classified as heavy, medium, or light purchasers:

First 3 Months' Purchases	Group (%)		
	Heavy	Medium	Light
0	5	15	60
1	10	30	20
2	30	40	15
3+	55	15	5

If a member purchases no books in the first 3 months, what is the probability that the member is a light purchaser? (*Note:* This table contains "conditional" percentages for each column.)

Solution Using the conditional probabilities in the table, the underlying purchase probabilities, and Bayes' Formula, we can compute this conditional probability.

$$\begin{aligned}
 P(\text{light}|0) &= \frac{P(0|\text{light})P(\text{light})}{P(0|\text{light})P(\text{light}) + P(0|\text{medium})P(\text{medium}) + P(0|\text{heavy})P(\text{heavy})} \\
 &= \frac{(.60)(.50)}{(.60)(.50) + (.15)(.30) + (.05)(.20)} \\
 &= .845
 \end{aligned}$$

states of nature
prior probabilities
observable events

likelihoods
posterior probabilities

These examples indicate the basic idea of Bayes' Formula. There is some number k of possible, mutually exclusive, underlying events $A_1, \dots, A_k$, which are sometimes called the **states of nature**. Unconditional probabilities $P(A_1), \dots, P(A_k)$, often called **prior probabilities**, are specified. There are m possible, mutually exclusive, **observable events** $B_1, \dots, B_m$. The conditional probabilities of each observable event given each state of nature, $P(B_i|A_j)$, are also specified, and these probabilities are called **likelihoods**. The problem is to find the **posterior probabilities** $P(A_i|B_j)$. *Prior* and *posterior* refer to probabilities before and after observing an event B_j .

Bayes' Formula

If $A_1, \dots, A_k$ are mutually exclusive states of nature, and if $B_1, \dots, B_m$ are m possible mutually exclusive observable events, then

$$\begin{aligned}
 P(A_i|B_j) &= \frac{P(B_j|A_i)P(A_i)}{P(B_j|A_1)P(A_1) + P(B_j|A_2)P(A_2) + \dots + P(B_j|A_k)P(A_k)} \\
 &= \frac{P(B_j|A_i)P(A_i)}{\sum_i P(B_j|A_i)P(A_i)}
 \end{aligned}$$

EXAMPLE 4.4

In the manufacture of circuit boards, there are three major types of defective boards. The types of defects, along with the percentage of all circuit boards having these defects, are (1) improper electrode coverage (D_1), 2.8%; (2) plating separation (D_2), 1.2%; and (3) etching problems (D_3), 3.2%. A circuit board will contain at most one of the three defects. Defects can be detected with certainty using destructive testing of the finished circuit boards; however, this is not a very practical method for inspecting a large percentage of the circuit boards. A nondestructive inspection procedure has been developed, which has the following outcomes: A_1 , which indicates the board has only defect D_1 ; A_2 , which indicates the board has only defect D_2 ; A_3 , which indicates the board has only defect D_3 ; and A_4 , which indicates the board has no defects. The respective likelihoods for the four outcomes of the nondestructive test determined by evaluating a large number of boards known to have exactly one of the three types of defects are given in the following table:

Test Outcome	Type of Defect			
	D_1	D_2	D_3	None
A_1	.90	.06	.02	.02
A_2	.05	.80	.06	.01
A_3	.03	.05	.82	.02
A_4 (no defects)	.02	.09	.10	.95

If a circuit board is tested using the nondestructive test and the outcome indicates no defects (A_4), what are the probabilities that the board has no defect or a D_1 , D_2 , or D_3 type of defect?

Let D_4 represent the situation in which the circuit board has no defects.

$$\begin{aligned}
 P(D_1|A_4) &= \frac{P(A_4|D_1)P(D_1)}{P(A_4|D_1)P(D_1) + P(A_4|D_2)P(D_2) + P(A_4|D_3)P(D_3) + P(A_4|D_4)P(D_4)} \\
 &= \frac{(.02)(.028)}{(.02)(.028) + (.09)(.012) + (.10)(.032) + (.95)(.928)} = \frac{.00056}{.88644} = .00063
 \end{aligned}$$

$$\begin{aligned}
 P(D_2|A_4) &= \frac{P(A_4|D_2)P(D_2)}{P(A_4|D_1)P(D_1) + P(A_4|D_2)P(D_2) + P(A_4|D_3)P(D_3) + P(A_4|D_4)P(D_4)} \\
 &= \frac{(.09)(.012)}{(.02)(.028) + (.09)(.012) + (.10)(.032) + (.95)(.928)} = \frac{.00108}{.88644} = .00122
 \end{aligned}$$

$$\begin{aligned}
 P(D_3|A_4) &= \frac{P(A_4|D_3)P(D_3)}{P(A_4|D_1)P(D_1) + P(A_4|D_2)P(D_2) + P(A_4|D_3)P(D_3) + P(A_4|D_4)P(D_4)} \\
 &= \frac{(.10)(.032)}{(.02)(.028) + (.09)(.012) + (.10)(.032) + (.95)(.928)} = \frac{.0032}{.88644} = .0036
 \end{aligned}$$

$$\begin{aligned}
 P(D_4|A_4) &= \frac{P(A_4|D_4)P(D_4)}{P(A_4|D_1)P(D_1) + P(A_4|D_2)P(D_2) + P(A_4|D_3)P(D_3) + P(A_4|D_4)P(D_4)} \\
 &= \frac{(.95)(.928)}{(.02)(.028) + (.09)(.012) + (.10)(.032) + (.95)(.928)} = \frac{.8816}{.88644} = .9945
 \end{aligned}$$

Thus, if the new test indicates that none of the three types of defects is present in the circuit board, there is a very high probability, .9945, that the circuit board in fact is free of defects. In Exercise 4.23, we will ask you to assess the sensitivity of the test for determining the three types of defects.

EXERCISES

Applications

Bus. 4.23 Of a finance company's loans, 1% are defaulted (not completely repaid). The company routinely runs credit checks on all loan applicants. It finds that 30% of defaulted loans went to poor risks, 40% to fair risks, and 30% to good risks. Of the nondefaulted loans, 10% went to poor risks, 40% to fair risks, and 50% to good risks. Use Bayes' Formula to calculate the probability that a poor-risk loan will be defaulted.

4.24 Refer to Exercise 4.23. Show that the posterior probability of default, given a fair risk, equals the prior probability of default. Explain why this is a reasonable result.

4.25 In Example 4.4, we described a new test for determining defects in circuit boards. Compute the probability that the test correctly identifies the defects D_1 , D_2 , and D_3 ; that is, compute $P(D_1|A_1)$, $P(D_2|A_2)$, and $P(D_3|A_3)$.

4.26 In Example 4.4, compute the probability that the test incorrectly identifies the defects D_1 , D_2 , and D_3 ; that is, compute $P(D_1|\bar{A}_1)$, $P(D_2|\bar{A}_2)$, and $P(D_3|\bar{A}_3)$.

Bus. 4.27 An underwriter of home insurance policies studies the problem of home fires resulting from wood-burning furnaces. Of all homes having such furnaces, 30% own a type 1 furnace, 25% a type 2 furnace, 15% a type 3, and 30% other types. Over 3 years, 5% of type 1 furnaces, 3% of type 2, 2% of type 3, and 4% of other types have resulted in fires. If a fire occurs in a particular home, what is the probability that a type 1 furnace is in the home?

Med. 4.28 In a January 15, 1998, article, the *New England Journal of Medicine* reported on the utility of using computerized tomography (CT) as a diagnostic test for patients with clinically suspected appendicitis. In at least 20% of patients with appendicitis, the correct diagnosis was not made. On the other hand, the appendix was normal in 15% to 40% of patients who underwent emergency appendectomy. A study was designed to determine the prospective effectiveness of using CT as a diagnostic test to improve the treatment of these patients. The study examined 100 consecutive patients suspected of having acute appendicitis who presented to the emergency department or were referred there from a physician's office. The 100 patients underwent a CT scan, and the surgeon made an assessment of the presence of appendicitis for each of the patients. The final clinical outcomes were determined at surgery and by pathological examination of the appendix after appendectomy or by clinical follow-up at least 2 months after CT scanning.

Radiologic Determination	Presence of Appendicitis	
	Confirmed (C)	Ruled Out (RO)
Definitely appendicitis (DA)	.94	.02
Equivocally appendicitis (EA)	.04	.04
Definitely not appendicitis (DNA)	.02	.94

The 1996 rate of occurrence of appendicitis was approximately $P(C) = .00108$.

- Find the sensitivity and specificity of the radiological determination of appendicitis.
- Find the probability that a patient truly had appendicitis given that the radiological determination was definite appendicitis (DA).
- Find the probability that a patient truly did not have appendicitis given that the radiological determination was definite appendicitis (DA).
- Find the probability that a patient truly did not have appendicitis given that the radiological determination was definitely not appendicitis (DNA).

Med. 4.29 Conditional probabilities can be useful in diagnosing disease. Suppose that three different, closely related diseases (A_1 , A_2 , and A_3) occur in 25%, 15%, and 12% of the population. In addition, suppose that any one of three mutually exclusive symptom states (B_1 , B_2 , and B_3) may be associated with each of these diseases. Experience shows that the likelihood $P(B_j|A_i)$ of having a given symptom state when the disease is present is as shown in the following table. Find the probability of disease A_2 given symptoms B_1 , B_2 , B_3 , and B_4 , respectively.

Symptom State B_j	Disease State A_i		
	A_1	A_2	A_3
B_1	.08	.17	.10
B_2	.18	.12	.14
B_3	.06	.07	.08
B_4 (no symptoms)	.68	.64	.68

4.6 Variables: Discrete and Continuous

The basic language of probability developed in this chapter deals with many different kinds of events. We are interested in calculating the probabilities associated with both quantitative and qualitative events. For example, we developed techniques that could be used to determine the probability that a machinist selected at random from the workers in a large automotive plant would suffer an accident during an 8-hour shift. These same techniques are also applicable to finding the probability that a machinist selected at random would work more than 80 hours without suffering an accident.

qualitative random variable

These qualitative and quantitative events can be classified as events (or outcomes) associated with qualitative and quantitative variables. For example, in the automotive accident study, the randomly selected machinist's accident report would consist of checking one of the following: No Accident, Minor Accident, or Major Accident. Thus, the data on 100 machinists in the study would be observations on a qualitative variable, because the possible responses are the different categories of accident and are not different in any measurable, numerical amount. Because we cannot predict with certainty what type of accident a particular machinist will suffer, the variable is classified as a **qualitative random variable**. Other examples of qualitative random variables that are commonly measured are political party affiliation, socioeconomic status, the species of insect discovered on an apple leaf, and the brand preferences of customers. There are a finite (and typically quite small) number of possible outcomes associated with any qualitative variable. Using the methods of this chapter, it is possible to calculate the probabilities associated with these events.

quantitative random variable

Many times the events of interest in an experiment are quantitative outcomes associated with a **quantitative random variable**, since the possible responses vary in numerical magnitude. For example, in the automotive accident study, the number of consecutive 8-hour shifts between accidents for a randomly selected machinist is an observation on a quantitative random variable. Events of interest, such as the number of 8-hour shifts between accidents for a randomly selected machinist, are observations on a quantitative random variable. Other examples of quantitative random variables are the change in earnings per share of a stock over the next quarter, the length of time a patient is in remission after a cancer treatment, the yield per acre of a new variety of wheat, and the number of persons voting for the incumbent in an upcoming election. The methods of this chapter can be applied to calculate the probability associated with any particular event.

random variable

There are major advantages to dealing with quantitative random variables. The numerical yardstick underlying a quantitative variable makes the mean and standard deviation (for instance) sensible. With qualitative random variables the methods of this chapter can be used to calculate the probabilities of various events, and that's about all. With quantitative random variables, we can do much more: we can average the resulting quantities, find standard deviations, and assess probable errors, among other things. Hereafter, we use the term **random variable** to mean quantitative random variable.

Most events of interest result in numerical observations or measurements. If a quantitative variable measured (or observed) in an experiment is denoted by the symbol y , we are interested in the values that y can assume. These values are called *numerical outcomes*. The number of different plant species per acre in a coal strip mine after a reclamation project is a numerical outcome. The percentage of registered voters who cast ballots in a given election is also a numerical outcome.

The quantitative variable y is called a *random variable* because the value that y assumes in a given experiment is a chance or random outcome.

DEFINITION 4.10

When observations on a quantitative random variable can assume only a countable number of values, the variable is called a **discrete random variable**.

Examples of discrete variables are these:

1. Number of bushels of apples per tree of a genetically altered apple variety
2. Change in the number of accidents per month at an intersection after a new signaling device has been installed
3. Number of “dead persons” voting in the last mayoral election in a major midwest city

Note that it is possible to count the number of values that each of these random variables can assume.

DEFINITION 4.11

When observations on a quantitative random variable can assume any one of the uncountable number of values in a line interval, the variable is called a **continuous random variable**.

For example, the daily maximum temperature in Rochester, New York, can assume any of the infinitely many values on a line interval. It can be 89.6, 89.799, or 89.7611114. Typical continuous random variables are temperature, pressure, height, weight, and distance.

The distinction between **discrete** and **continuous random** variables is pertinent when we are seeking the probabilities associated with specific values of a random variable. The need for the distinction will be apparent when probability distributions are discussed in later sections of this chapter.

4.7**Probability Distributions for Discrete Random Variables****probability distribution**

As previously stated, we need to know the probability of observing a particular sample outcome in order to make an inference about the population from which the sample was drawn. To do this, we need to know the probability associated with each value of the variable y . Viewed as relative frequencies, these probabilities generate a distribution of theoretical relative frequencies called the **probability distribution** of y . Probability distributions differ for discrete and continuous random variables. For discrete random variables, we will compute the probability of specific individual values occurring. For continuous random variables, the probability of an interval of values is the event of interest.

The *probability distribution for a discrete random variable* displays the probability $P(y)$ associated with each value of y . This display can be presented as a table, a graph, or a formula. To illustrate, consider the tossing of two coins in

Section 4.2 and let y be the number of heads observed. Then y can take the values 0, 1, or 2. From the data of Table 4.1, we can determine the approximate probability for each value of y , as given in Table 4.3. We point out that the relative frequencies in the table are very close to the theoretical relative frequencies (probabilities), which can be shown to be .25, .50, and .25 using the classical interpretation of probability. If we had employed 2,000,000 tosses of the coins instead of 500, the relative frequencies for $y = 0, 1$, and 2 would be indistinguishable from the theoretical probabilities.

TABLE 4.3

Empirical sampling results for y : the number of heads in 500 tosses of two coins

y	Frequency	Relative Frequency
0	129	.258
1	242	.484
2	129	.258

The probability distribution for y , the number of heads in the toss of two coins, is shown in Table 4.4 and is presented graphically as a *probability histogram* in Figure 4.2.

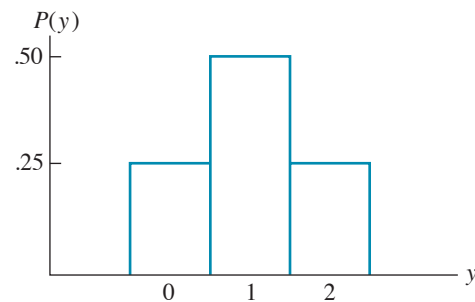
TABLE 4.4

Probability distribution for the number of heads when two coins are tossed

y	$P(y)$
0	.25
1	.50
2	.25

FIGURE 4.2

Probability distribution for the number of heads when two coins are tossed



The probability distribution for this simple discrete random variable illustrates three important properties of discrete random variables.

Properties of Discrete Random Variables

1. The probability associated with every value of y lies between 0 and 1.
2. The sum of the probabilities for all values of y is equal to 1.
3. The probabilities for a discrete random variable are additive. Hence, the probability that $y = 1$ or 2 is equal to $P(1) + P(2)$.

The relevance of the probability distribution to statistical inference will be emphasized when we discuss the probability distribution for the binomial random variable.

4.8 A Useful Discrete Random Variable: The Binomial

Many populations of interest to business persons and scientists can be viewed as large sets of 0s and 1s. For example, consider the set of responses of all adults in the United States to the question, “Do you favor the development of nuclear energy?” If we disallow “no opinion,” the responses will constitute a set of “yes” responses and “no” responses. If we assign a 1 to each yes and a 0 to each no, the population will consist of a set of 0s and 1s, and the sum of the 1s will equal the total number of persons favoring the development. The sum of the 1s divided by the number of adults in the United States will equal the proportion of people who favor the development.

Gallup and Harris polls are examples of the sampling of 0, 1 populations. People are surveyed, and their opinions are recorded. Based on the sample responses, Gallup and Harris estimate the proportions of people in the population who favor some particular issue or possess some particular characteristic.

Similar surveys are conducted in the biological sciences, engineering, and business, but they may be called experiments rather than polls. For example, experiments are conducted to determine the effect of new drugs on small animals, such as rats or mice, before progressing to larger animals and, eventually, to human participants. Many of these experiments bear a marked resemblance to a poll in that the experimenter records only whether the drug was effective. Thus, if 300 rats are injected with a drug and 230 show a favorable response, the experimenter has conducted a “poll”—a poll of rat reaction to the drug, 230 “in favor” and 70 “opposed.”

Similar “polls” are conducted by most manufacturers to determine the fraction of a product that is of good quality. Samples of industrial products are collected before shipment and each item in the sample is judged “defective” or “acceptable” according to criteria established by the company’s quality control department. Based on the number of defectives in the sample, the company can decide whether the product is suitable for shipment. Note that this example, as well as those preceding, has the practical objective of making an inference about a population based on information contained in a sample.

The public opinion poll, the consumer preference poll, the drug-testing experiment, and the industrial sampling for defectives are all examples of a common, frequently conducted sampling situation known as a *binomial experiment*. The binomial experiment is conducted in all areas of science and business and only differs from one situation to another in the nature of objects being sampled (people, rats, electric lightbulbs, oranges). Thus, it is useful to define its characteristics. We can then apply our knowledge of this one kind of experiment to a variety of sampling experiments.

For all practical purposes the binomial experiment is identical to the coin-tossing example of previous sections. Here, n different coins are tossed (or a single coin is tossed n times), and we are interested in the number of heads observed. We assume that the probability of tossing a head on a single trial is π (π may equal .50, as it would for a balanced coin, but in many practical situations π will take some other value between 0 and 1). We also assume that the outcome for any one toss is unaffected by the results of any preceding tosses. These characteristics can be summarized as shown here.

DEFINITION 4.12

A **binomial experiment** is one that has the following properties:

1. The experiment consists of n identical trials.
2. Each trial results in one of two outcomes. We will label one outcome a success and the other a failure.
3. The probability of success on a single trial is equal to π and π remains the same from trial to trial.*
4. The trials are independent; that is, the outcome of one trial does not influence the outcome of any other trial.
5. The random variable y is the number of successes observed during the n trials.

EXAMPLE 4.5

An article in the March 5, 1998, issue of *The New England Journal of Medicine* discussed a large outbreak of tuberculosis. One person, called the index patient, was diagnosed with tuberculosis in 1995. The 232 co-workers of the index patient were given a tuberculin screening test. The number of co-workers recording a positive reading on the test was the random variable of interest. Did this study satisfy the properties of a binomial experiment?

Solution To answer the question, we check each of the five characteristics of the binomial experiment to determine whether they were satisfied.

1. Were there n identical trials? Yes. There were $n = 232$ workers who had approximately equal contact with the index patient.
2. Did each trial result in one of two outcomes? Yes. Each co-worker recorded either a positive or negative reading on the test.
3. Was the probability of success the same from trial to trial? Yes, if the co-workers had equivalent risk factors and equal exposures to the index patient.
4. Were the trials independent? Yes. The outcome of one screening test was unaffected by the outcome of the other screening tests.
5. Was the random variable of interest to the experimenter the number of successes y in the 232 screening tests? Yes. The number of co-workers who obtained a positive reading on the screening test was the variable of interest.

All five characteristics were satisfied, so the tuberculin screening test represented a binomial experiment.

EXAMPLE 4.6

An economist interviews 75 students in a class of 100 to estimate the proportion of students who expect to obtain a “C” or better in the course. Is this a binomial experiment?

* Some textbooks and computer programs use the letter p rather than π . We have chosen π to avoid confusion with p -values, discussed in Chapter 5.

Solution Check this experiment against the five characteristics of a binomial experiment.

1. Are there identical trials? Yes. Each of 75 students is interviewed.
2. Does each trial result in one of two outcomes? Yes. Each student either does or does not expect to obtain a grade of “C” or higher.
3. Is the probability of success the same from trial to trial? No. If we let success denote a student expecting to obtain a “C” or higher, then the probability of success can change considerably from trial to trial. For example, unknown to the professor, suppose that 75 of the 100 students expect to obtain a grade of “C” or higher. Then π , the probability of success for the first student interviewed, is $75/100 = .75$. If the student is a failure (does not expect a “C” or higher), the probability of success for the next student is $75/99 = .76$. Suppose that after 70 students are interviewed, 60 are successes and 10 are failures. Then the probability of success for the next (71st) student is $15/30 = .50$.

This example shows how the probability of success can change substantially from trial to trial in situations in which the sample size is a relatively large portion of the total population size. This experiment does not satisfy the properties of a binomial experiment.

Note that very few real-life situations satisfy perfectly the requirements stated in Definition 4.12, but for many the lack of agreement is so small that the binomial experiment still provides a very good model for reality.

Having defined the binomial experiment and suggested several practical applications, we now examine the probability distribution for the binomial random variable y , the number of successes observed in n trials. Although it is possible to approximate $P(y)$, the probability associated with a value of y in a binomial experiment, by using a relative frequency approach, it is easier to use a general formula for binomial probabilities.

Formula for Computing $P(y)$ in a Binomial Experiment

The probability of observing y successes in n trials of a binomial experiment is

$$P(y) = \frac{n!}{y!(n-y)!} \pi^y (1-\pi)^{n-y}$$

where

n = number of trials

π = probability of success on a single trial

$1 - \pi$ = probability of failure on a single trial

y = number of successes in n trials

$n! = n(n-1)(n-2) \cdots (3)(2)(1)$

As indicated in the box, the notation $n!$ (referred to as n factorial) is used for the product

$$n! = n(n-1)(n-2) \cdots (3)(2)(1)$$

For $n = 3$,

$$n! = 3! = (3)(3 - 1)(3 - 2) = (3)(2)(1) = 6$$

Similarly, for $n = 4$,

$$4! = (4)(3)(2)(1) = 24$$

We also note that $0!$ is defined to be equal to 1.

To see how the formula for binomial probabilities can be used to calculate the probability for a specific value of y , consider the following examples.

EXAMPLE 4.7

A new variety of turf grass has been developed for use on golf courses, with the goal of obtaining a germination rate of 85%. To evaluate the grass, 20 seeds are planted in a greenhouse so that each seed will be exposed to identical conditions. If the 85% germination rate is correct, what is the probability that 18 or more of the 20 seeds will germinate?

$$P(y) = \frac{n!}{y!(n - y)!} \pi^y (1 - \pi)^{n-y}$$

and substituting for $n = 20$, $\pi = .85$, $y = 18, 19$, and 20 , we obtain

$$P(y = 18) = \frac{20!}{18!(20 - 18)!} (.85)^{18} (1 - .85)^{20-18} = 190(.85)^{18} (.15)^2 = .229$$

$$P(y = 19) = \frac{20!}{19!(20 - 19)!} (.85)^{19} (1 - .85)^{20-19} = 20(.85)^{19} (.15)^1 = .137$$

$$P(y = 20) = \frac{20!}{20!(20 - 20)!} (.85)^{20} (1 - .85)^{20-20} = (.85)^{20} = .0388$$

$$P(y \geq 18) = P(y = 18) + P(y = 19) + P(y = 20) = .405$$

The calculations in Example 4.7 entail a considerable amount of effort even though n was only 20. For those situations involving a large value of n , we can use computer software to make the exact calculations. An approach that yields fairly accurate results in many situations and does not require the use of a computer will be discussed later in this chapter.

EXAMPLE 4.8

Suppose that a sample of households is randomly selected from all the households in the city in order to estimate the percentage in which the head of the household is unemployed. To illustrate the computation of a binomial probability, suppose that the unknown percentage is actually 10% and that a sample of $n = 5$ (we select a small sample to make the calculation manageable) is selected from the population. What is the probability that all five heads of the households are employed?

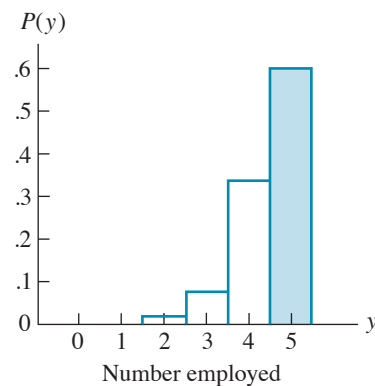
Solution We must carefully define which outcome we wish to call a success. For this example, we define a success as being employed. Then the probability of success when one person is selected from the population is $\pi = .9$ (because the

proportion unemployed is .1). We wish to find the probability that $y = 5$ (all five are employed) in five trials.

$$\begin{aligned} P(y = 5) &= \frac{5!}{5!(5-5)!} (.9)^5 (.1)^0 \\ &= \frac{5!}{5!0!} (.9)^5 (.1)^0 \\ &= (.9)^5 = .590 \end{aligned}$$

The binomial probability distribution for $n = 5$, $\pi = .9$ is shown in Figure 4.3. The probability of observing five employed in a sample of five is shaded in the figure.

FIGURE 4.3
The binomial probability distribution for $n = 5$, $\pi = .9$



EXAMPLE 4.9

Refer to Example 4.8 and calculate the probability that exactly one person in the sample of five households is unemployed. What is the probability of one or fewer being unemployed?

Solution Since y is the number of employed in the sample of five, one unemployed person would correspond to four employed ($y = 4$). Then

$$\begin{aligned} P(4) &= \frac{5!}{4!(5-4)!} (.9)^4 (.1)^1 \\ &= \frac{(5)(4)(3)(2)(1)}{(4)(3)(2)(1)(1)} (.9)^4 (.1) \\ &= 5(.9)^4 (.1) \\ &= .328 \end{aligned}$$

Thus, the probability of selecting four employed heads of households in a sample of five is .328, or, roughly, one chance in three.

The outcome “one or fewer unemployed” is the same as the outcome “4 or 5 employed.” Since y represents the number employed, we seek the probability that $y = 4$ or 5. Because the values associated with a random variable represent mutually exclusive events, the probabilities for discrete random variables are

additive. Thus, we have

$$\begin{aligned} P(y = 4 \text{ or } 5) &= P(4) + P(5) \\ &= .328 + .590 \\ &= .918 \end{aligned}$$

Thus, the probability that a random sample of five households will yield either four or five employed heads of households is .918. This high probability is consistent with our intuition: we could expect the number of employed in the sample to be large if 90% of all heads of households in the city are employed.

Like any relative frequency histogram, a binomial probability distribution possesses a mean, μ , and a standard deviation, σ . Although we omit the derivations, we give the formulas for these parameters.

**Mean and Standard
Deviation of the Binomial
Probability Distribution**

$$\mu = n\pi \quad \text{and} \quad \sigma = \sqrt{n\pi(1 - \pi)}$$

where π is the probability of success in a given trial and n is the number of trials in the binomial experiment.

If we know π and the sample size, n , we can calculate μ and σ to locate the center and describe the variability for a particular binomial probability distribution. Thus, we can quickly determine those values of y that are probable and those that are improbable.

EXAMPLE 4.10

We will consider the turf grass seed example to illustrate the calculation of the mean and standard deviation. Suppose the company producing the turf grass takes a sample of 20 seeds on a regular basis to monitor the quality of the seeds. If the germination rate of the seeds stays constant at 85%, then the average number of seeds that will germinate in the sample of 20 seeds is

$$\mu = n\pi = 20(.85) = 17$$

with a standard deviation of

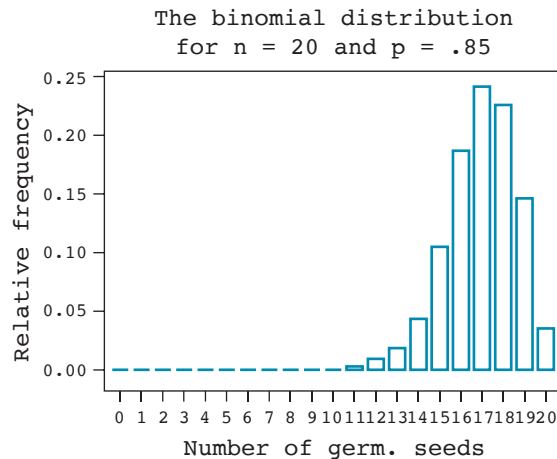
$$\sigma = \sqrt{n\pi(1 - \pi)} = \sqrt{20(.85)(1 - .85)} = 1.60$$

Suppose we examine the germination records of a large number of samples of 20 seeds each. If the germination rate has remained constant at 85%, then the average number of seeds that germinate should be close to 17 per sample. If in a particular sample of 20 seeds we determine that only 12 had germinated, would the germination rate of 85% seem consistent with our results? Using a computer software program, we can generate the probability distribution for the number of seeds that germinate in the sample of 20 seeds, as shown in Figure 4.4.

Although the distribution is tending toward left skewness (see Figure 4.4), the Empirical Rule should work well for this relatively mound-shaped distribution. Thus, $y = 12$ seeds is more than 3 standard deviations less than the mean number of seeds, $\mu = 17$; it is highly improbable that in 20 seeds we would obtain only 12 germinated seeds if π really is equal to .85. The germination rate is most likely a value considerably less than .85.

FIGURE 4.4

The binomial distribution for
 $n = 20$ and $p = .85$

**EXAMPLE 4.11**

A poll shows that 516 of 1,218 voters favor the reelection of a particular political candidate. Do you think that the candidate will win?

Solution To win the election, the candidate will need at least 50% of the votes. Let us see whether $y = 516$ is too small a value of y to imply a value of π (the proportion of voters favoring the candidate) equal to .5 or larger. If $\pi = .5$,

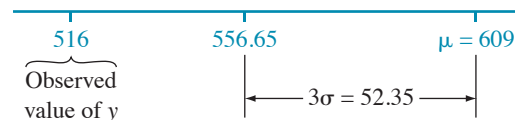
$$\begin{aligned}\mu &= n\pi = (1,218)(.5) = 609 \\ \sigma &= \sqrt{n\pi(1 - \pi)} = \sqrt{(1,218)(.5)(.5)} \\ &= \sqrt{304.5} = 17.45\end{aligned}$$

and $3\sigma = 52.35$.

You can see from Figure 4.5 that $y = 516$ is more than 3σ , or 52.35, away from $\mu = 609$. In fact, if you wish to check, you will see that $y = 516$ is more than 5σ away from $\mu = 609$, the value of μ if π were really equal to .5. Thus, it appears that the number of voters in the sample who favor the candidate is much too small if the candidate does, in fact, possess a majority favoring reelection. Consequently, we conclude that he or she will lose. (Note that this conclusion is based on the assumption that the set of voters from which the sample was drawn is the same as the set who will vote. We also must assume that the opinions of the voters will not change between the time of sampling and the date of the election.)

FIGURE 4.5

Location of the observed
value of y ($y = 516$) relative
to μ



The purpose of this section is to present the binomial probability distribution so you can see how binomial probabilities are calculated and so you can calculate them for small values of n , if you wish. In practice, n is usually large (in national surveys, sample sizes as large as 1,500 are common), and the computation of the binomial probabilities is tedious. Later in this chapter, we will present a simple

procedure for obtaining approximate values to the probabilities we need in making inferences. We can also use some very rough procedures for evaluating probabilities by using the mean and standard deviation of the binomial random variable y along with the Empirical Rule.

The only other discrete random variable that we will discuss in this text is the Poisson (Chapter 10). We refer the interested reader to Hildebrand and Ott (1998) and Devore (2000) for more information about discrete random variables. In the next section, we discuss probability distributions with emphasis on the normal distribution.

EXERCISES**Basic Techniques**

4.30 Consider the following class experiment: Toss three coins and observe the number of heads y . Let each student repeat the experiment ten times, combine the class results, and construct a relative frequency table for y . Note that these frequencies give approximations to the actual probabilities that $y = 0, 1, 2$, or 3 . (*Note:* Calculate the actual probabilities by using the binomial formula $P(y)$ to compare the approximate results with the actual probabilities.)

4.31 Let y be a binomial random variable; compute $P(y)$ for each of the following situations:

- a. $n = 10, \pi = .2, y = 3$
- b. $n = 4, \pi = .4, y = 2$
- c. $n = 16, \pi = .7, y = 12$

4.32 Let y be a binomial random variable with $n = 8$ and $\pi = .4$. Find the following values:

- a. $P(y \leq 4)$
- b. $P(y > 4)$
- c. $P(y \leq 7)$
- d. $P(y > 6)$

Applications**Bus.**

4.33 An appliance store has the following probabilities for y , the number of major appliances sold on a given day:

y	$P(y)$
0	.100
1	.150
2	.250
3	.140
4	.090
5	.080
6	.060
7	.050
8	.040
9	.025
10	.015

- a. Construct a graph of $P(y)$.
- b. Find $P(y \leq 2)$.
- c. Find $P(y \geq 7)$.
- d. Find $P(1 \leq y \leq 5)$.

- Bus. 4.34** The weekly demand for copies of a popular word-processing program at a computer store has the probability distribution shown here.

y	$P(y)$
0	.06
1	.14
2	.16
3	.14
4	.12
5	.10
6	.08
7	.07
8	.06
9	.04
10	.03

- What is the probability that three or more copies will be needed in a particular week?
 - What is the probability that the demand will be for at least two but no more than six copies?
 - If the store has eight copies of the program available at the beginning of each week, what is the probability the demand will exceed the supply in a given week?
- Bio. 4.35** A biologist randomly selects ten portions of water, each equal to $.1 \text{ cm}^3$ in volume, from the local reservoir and counts the number of bacteria present in each portion. The biologist then totals the number of bacteria for the ten portions to obtain an estimate of the number of bacteria per cubic centimeter present in the reservoir water. Is this a binomial experiment?
- Pol. Sci. 4.36** Examine the accompanying newspaper clipping. Does this sampling appear to satisfy the characteristics of a binomial experiment?

Poll Finds Opposition to Phone Taps

New York—People surveyed in a recent poll indicated they are 81% to 13% against having their phones tapped without a court order.

The people in the survey, by 68% to 27%, were opposed to letting the government use a wiretap on citizens suspected of crimes, except with a court order.

The survey was conducted for 1,495 households and also found the following results:

—The people surveyed are 80% to

12% against the use of any kind of electronic spying device without a court order.

—Citizens are 77% to 14% against allowing the government to open their mail without court orders.

—They oppose, by 80% to 12%, letting the telephone company disclose records of long-distance phone calls, except by court order.

For each of the questions, a few of those in the survey had no responses.

- Env. 4.37** A survey is conducted to estimate the percentage of pine trees in a forest that are infected by the pine shoot moth. A grid is placed over a map of the forest, dividing the area into 25-foot by 25-foot square sections. One hundred of the squares are randomly selected and the number of infected trees is recorded for each square. Is this a binomial experiment?

Pol. Sci. **4.38** A survey was conducted to investigate the attitudes of nurses working in Veterans Administration hospitals. A sample of 1,000 nurses was contacted using a mailed questionnaire and the number favoring or opposing a particular issue was recorded. If we confine our attention to the nurses' responses to a single question, would this sampling represent a binomial experiment? As with most mail surveys, some of the nurses did not respond. What effect might nonresponses in the sample have on the estimate of the percentage of all Veterans Administration nurses who favor the particular proposition?

Env. **4.39** In an inspection of automobiles in Los Angeles, 60% of all automobiles had emissions that do not meet EPA regulations. For a random sample of ten automobiles, compute the following probabilities:

- All ten automobiles failed the inspection.
- Exactly six of the ten failed the inspection.
- Six or more failed the inspection.
- All ten passed the inspection.

Use the following Minitab output to answer the questions. Note that with Minitab, the binomial probability π is denoted by p and the binomial variable y is represented by x .

Binomial Distribution with $n = 10$ and $p = 0.6$

x	$P(X = x)$	$P(X \leq x)$
0.00	0.0001	0.0001
1.00	0.0016	0.0017
2.00	0.0106	0.0123
3.00	0.0425	0.0548
4.00	0.1115	0.1662
5.00	0.2007	0.3669
6.00	0.2508	0.6177
7.00	0.2150	0.8327
8.00	0.1209	0.9536
9.00	0.0403	0.9940
10.00	0.0060	1.0000

4.40 Refer to Exercise 4.39.

- Compute the probabilities for parts (a) through (d) if $\pi = .3$.
- Indicate how you would compute $P(y \leq 100)$ for $n = 1,000$ and $\pi = .3$.

Bio. **4.41** An experiment is conducted to test the effect of an anticoagulant drug on rats. A random sample of four rats is employed in the experiment. If the drug manufacturer claims that 80% of the rats will be favorably affected by the drug, what is the probability that none of the four experimental rats will be favorably affected? One of the four? One or fewer?

Soc. **4.42** A criminologist claims that the probability of reform for a first-offender embezzler is .9. Suppose that we define *reform* as meaning the person commits no criminal offenses within a 5-year period. Three paroled embezzlers were randomly selected from the prison records, and their behavioral histories were examined for the 5-year period following prison release. If the criminologist's claim is correct, what is the probability that all three were reformed? At least two?

4.43 Consider the following experiment: Toss three coins and observe the number of heads y . Repeat the experiment 100 times and construct a relative frequency table for y . Note that these frequencies give approximations to the exact probabilities that $y = 0, 1, 2$, and 3. (Note: These probabilities can be shown to be $1/8, 3/8, 3/8$, and $1/8$, respectively.)

4.44 Refer to Exercise 4.43. Use the formula for the binomial probability distribution to show that $P(0) = 1/8, P(1) = 3/8, P(2) = 3/8$, and $P(3) = 1/8$.

4.45 Suppose you match coins with another person a total of 1,000 times. What is the mean number of matches? The standard deviation? Calculate the interval $(\mu \pm 3\sigma)$. (*Hint:* The probability of a match in the toss of a single pair of coins is $\pi = .5$.)

4.46 Refer to Exercise 4.39. Indicate how you could compute $P(y \leq 100)$ if $n = 1,000$ for $\pi = .6$.

Bus. 4.47 Over a long period of time in a large multinational corporation, 10% of all sales trainees are rated as outstanding, 75% are rated as excellent/good, 10% are rated as satisfactory, and 5% are considered unsatisfactory. Find the following probabilities for a sample of ten trainees selected at random:

- Two are rated as outstanding.
- Two or more are rated as outstanding.
- Eight of the ten are rated either outstanding or excellent/good.
- None of the trainees is rated as unsatisfactory.

Med. 4.48 A new technique, balloon angioplasty, is being widely used to open clogged heart valves and vessels. The balloon is inserted via a catheter and is inflated, opening the vessel; thus, no surgery is required. Left untreated, 50% of the people with heart-valve disease die within about 2 years. If experience with this technique suggests that approximately 70% live for more than 2 years, would the next five patients of the patients treated with balloon angioplasty at a hospital constitute a binomial experiment with $n = 5$, $\pi = .70$? Why or why not?

Med. 4.49 A prescription drug firm claims that only 12% of all new drugs shown to be effective in animal tests ever make it through a clinical testing program and onto the market. If a firm has 15 new compounds that have shown effectiveness in animal tests, find the following probabilities:

- None reach the market.
- One or more reach the market.
- Two or more reach the market.

4.50 Does Exercise 4.49 satisfy the properties of a binomial experiment? Why or why not?

Bus. 4.51 A random sample of 50 price changes is selected from the many listed for a large supermarket during a reporting period. If the probability that a price change is posted correctly is .93,

- Write an expression for the probability that three or fewer changes are posted incorrectly.
- What assumptions were made for part (a)?

4.9 Probability Distributions for Continuous Random Variables

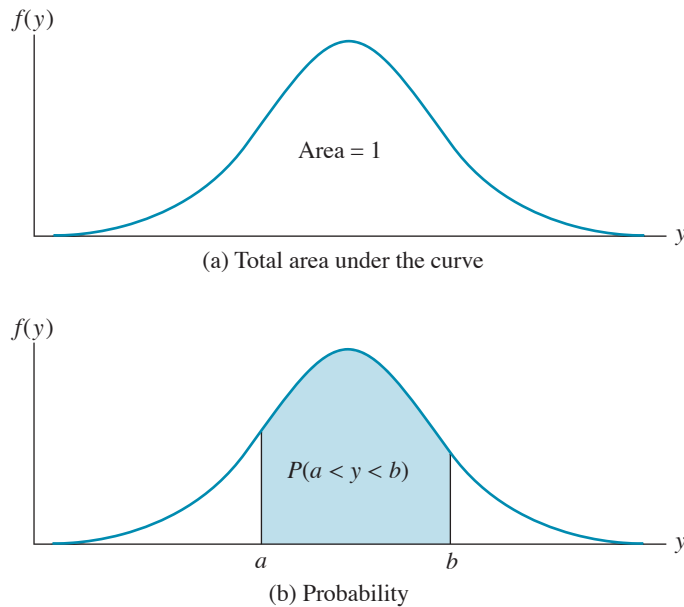
Discrete random variables (such as the binomial) have possible values that are distinct and separate, such as 0 or 1 or 2 or 3. Other random variables are most usefully considered to be *continuous*: their possible values form a whole interval (or range, or continuum). For instance, the 1-year return per dollar invested in a common stock could range from 0 to some quite large value. In practice, virtually all random variables assume a discrete set of values; the return per dollar of a million-dollar common-stock investment could be \$1.06219423 or \$1.06219424 or \$1.06219425 or However, when there are many possible values for a random variable, it is sometimes mathematically useful to treat the random variable as continuous.

Theoretically, then, a continuous random variable is one that can assume values associated with infinitely many points in a line interval. We state, without elaboration, that it is impossible to assign a small amount of probability to each

value of y (as was done for a discrete random variable) and retain the property that the probabilities sum to 1.

To overcome this difficulty, we revert to the concept of the relative frequency histogram of Chapter 3, where we talked about the probability of y falling in a given interval. Recall that the relative frequency histogram for a population containing a large number of measurements will almost be a smooth curve because the number of class intervals can be made large and the width of the intervals can be decreased. Thus, we envision a smooth curve that provides a model for the population relative frequency distribution generated by repeated observation of a continuous random variable. This will be similar to the curve shown in Figure 4.6.

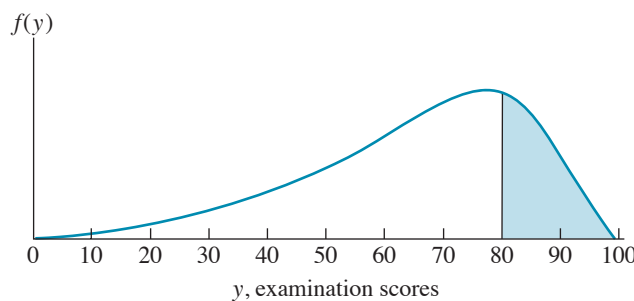
FIGURE 4.6
Probability distribution for a continuous random variable



Recall that the histogram relative frequencies are proportional to areas over the class intervals and that these areas possess a probabilistic interpretation. Thus, if a measurement is randomly selected from the set, the probability that it will fall in an interval is proportional to the histogram area above the interval. Since a population is the whole (100%, or 1), we want the total area under the probability curve to equal 1. If we let the total area under the curve equal 1, then areas over intervals are exactly equal to the corresponding probabilities.

The graph for the probability distribution for a continuous random variable is shown in Figure 4.7. The ordinate (height of the curve) for a given value of y is denoted by the symbol $f(y)$. Many people are tempted to say that $f(y)$, like $P(y)$ for the binomial random variable, designates the probability associated with the

FIGURE 4.7
Hypothetical probability distribution for student examination scores



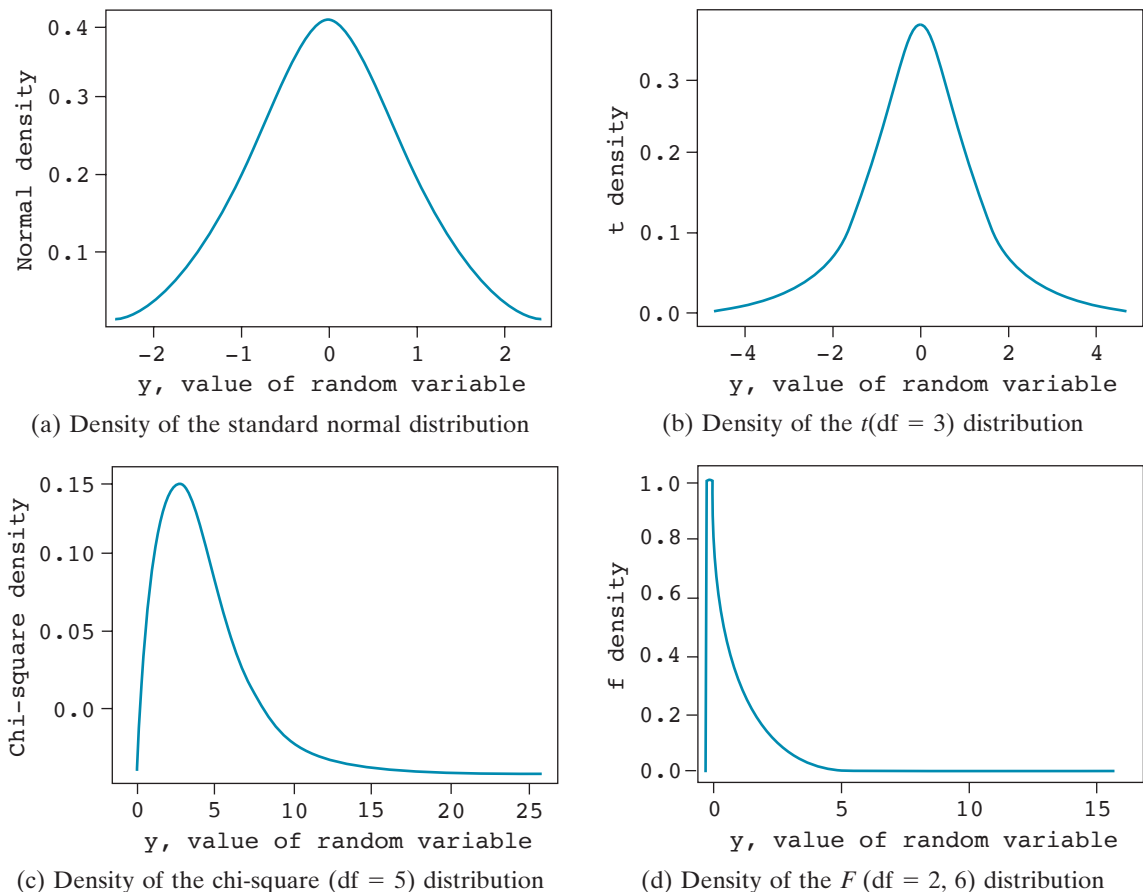
continuous random variable y . However, as we mentioned before, it is impossible to assign a probability to each of the infinitely many possible values of a continuous random variable. Thus, all we can say is that $f(y)$ represents the height of the probability distribution for a given value of y .

The probability that a continuous random variable falls in an interval, say between two points a and b , follows directly from the probabilistic interpretation given to the area over an interval for the relative frequency histogram (Section 3.3) and is equal to the area under the curve over the interval a to b , as shown in Figure 4.6. This probability is written $P(a < y < b)$.

There are curves of many shapes that can be used to represent the population relative frequency distribution for measurements associated with a continuous random variable. Fortunately, the areas for many of these curves have been tabulated and are ready for use. Thus, if we know that student examination scores possess a particular probability distribution, as in Figure 4.7, and if areas under the curve have been tabulated, we can find the probability that a particular student will score more than 80% by looking up the tabulated area, which is shaded in Figure 4.7.

Figure 4.8 depicts four important probability distributions that will be used extensively in the following chapters. Which probability distribution we use in a particular situation is very important because probability statements are determined by the area under the curve. As can be seen in Figure 4.8, we would obtain

FIGURE 4.8



very different answers depending on which distribution is selected. For example, the probability the random variable takes on a value less than 5.0 is essentially 1.0 for the probability distributions in Figures 4.8(a) and (b) but is 0.584 and 0.947 for the probability distributions in Figures 4.8(c) and (d), respectively. In some situations, we will not know exactly the distribution for the random variable in a particular study. In these situations, we can use the observed values for the random variable to construct a relative frequency histogram, which is a sample estimate of the true probability frequency distribution. As far as statistical inferences are concerned, the selection of the *exact* shape of the probability distribution for a continuous random variable is not crucial in many cases, because most of our inference procedures are insensitive to the exact specification of the shape.

We will find that data collected on continuous variables often possess a nearly bell-shaped frequency distribution, such as depicted in Figure 4.8(a). A continuous variable (the normal) and its probability distribution (bell-shaped curve) provide a good model for these types of data. The normally distributed variable is also very important in statistical inference. We will study the normal distribution in detail in the next section.

4.10 A Useful Continuous Random Variable: The Normal Distribution

normal curve

Many variables of interest, including several statistics to be discussed in later sections and chapters, have mound-shaped frequency distributions that can be approximated by using a **normal curve**. For example, the distribution of total scores on the Brief Psychiatric Rating Scale for outpatients having a current history of repeated aggressive acts is mound-shaped. Other practical examples of mound-shaped distributions are social perceptiveness scores of preschool children selected from a particular socioeconomic background, psychomotor retardation scores for patients with circular-type manic-depressive illness, milk yields for cattle of a particular breed, and perceived anxiety scores for residents of a community. Each of these mound-shaped distributions can be approximated with a normal curve.

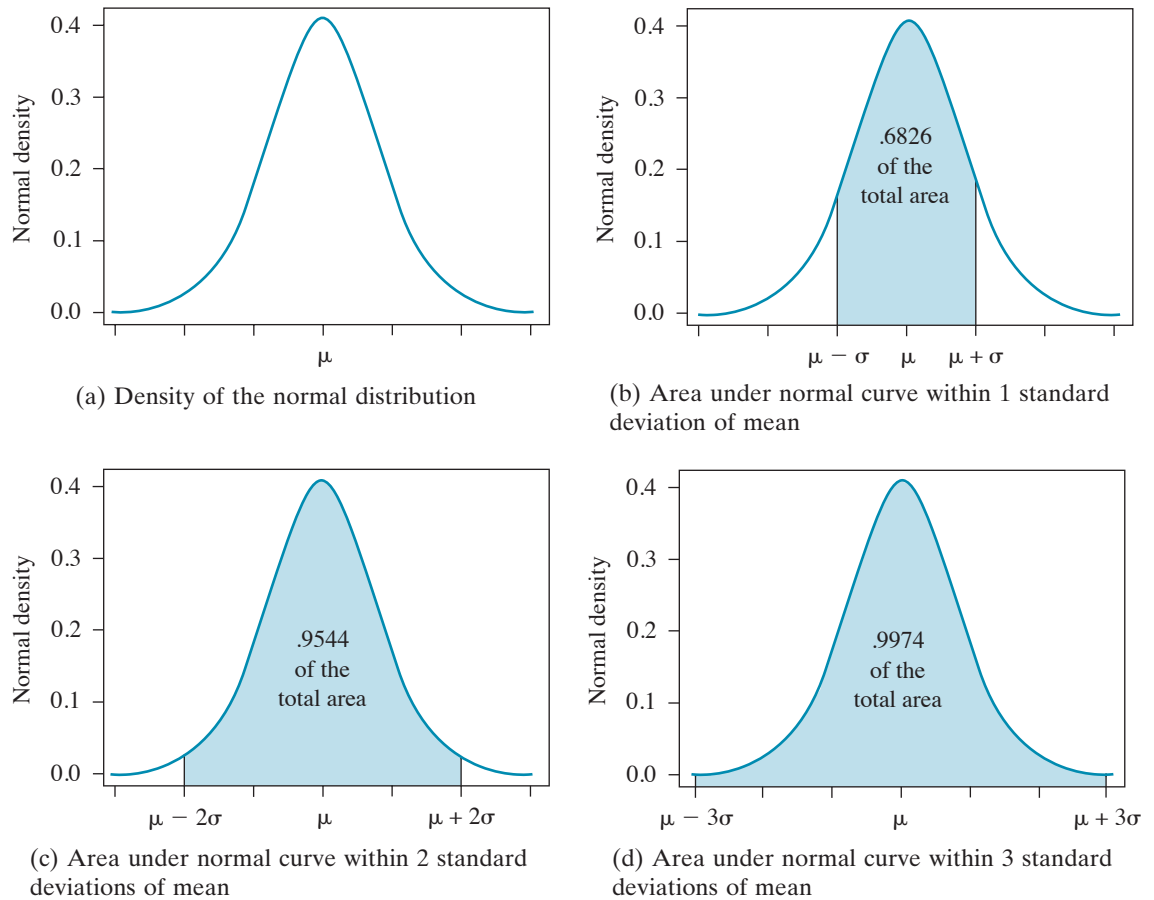
Since the normal distribution has been well tabulated, areas under a normal curve—which correspond to probabilities—can be used to approximate probabilities associated with the variables of interest in our experimentation. Thus, the normal random variable and its associated distribution play an important role in statistical inference.

The relative frequency histogram for the normal random variable, called the *normal curve* or *normal probability distribution*, is a smooth bell-shaped curve. Figure 4.9(a) shows a normal curve. If we let y represent the normal random variable, then the height of the probability distribution for a specific value of y is represented by $f(y)$.^{*} The probabilities associated with a normal curve form the basis for the Empirical Rule.

As we see from Figure 4.9(a), the normal probability distribution is bell shaped and symmetrical about the mean μ . Although the normal random variable y may theoretically assume values from $-\infty$ to $+\infty$, we know from the Empirical

^{*} For the normal distribution, $f(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(y-\mu)^2/2\sigma^2}$, where μ and σ are the mean and standard deviation, respectively, of the population of y -values.

FIGURE 4.9



Rule that approximately all the measurements are within 3 standard deviations (3σ) of μ . From the Empirical Rule, we also know that if we select a measurement at random from a population of measurements that possesses a mound-shaped distribution, the probability is approximately .68 that the measurement will lie within 1 standard deviation of its mean (see Figure 4.9(b)). Similarly, we know that the probability is approximately .954 that a value will lie in the interval $\mu \pm 2\sigma$ and .997 in the interval $\mu \pm 3\sigma$ (see Figures 4.9(c) and (d)). What we do not know, however, is the probability that the measurement will be within 1.65 standard deviations of its mean, or within 2.58 standard deviations of its mean. The procedure we are going to discuss in this section will enable us to calculate the probability that a measurement falls within any distance of the mean μ for a normal curve.

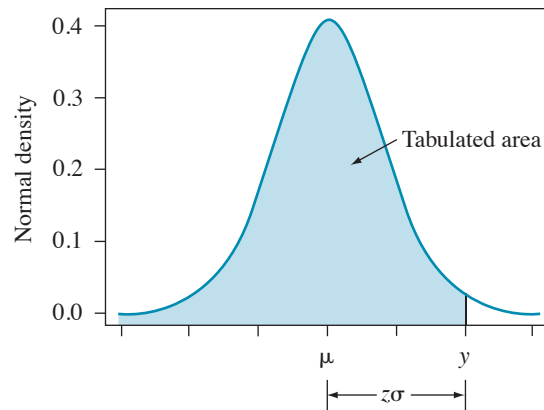
Because there are many different normal curves (depending on the parameters μ and σ), it might seem to be an impossible task to tabulate areas (probabilities) for all normal curves, especially if each curve requires a separate table. Fortunately, this is not the case. By specifying the probability that a variable y lies within a certain number of standard deviations of its mean (just as we did in using the Empirical Rule), we need only one table of probabilities.

area under a normal curve

Table 1 in the Appendix gives the **area under a normal curve** to the left of a value y that is z standard deviations ($z\sigma$) away from the mean (see Figure 4.10). The area shown by the shading in Figure 4.10 is the probability listed in Table 1 in the Appendix. Values of z to the nearest tenth are listed along the left-hand column of the table, with z to the nearest hundredth along the top of the table.

FIGURE 4.10

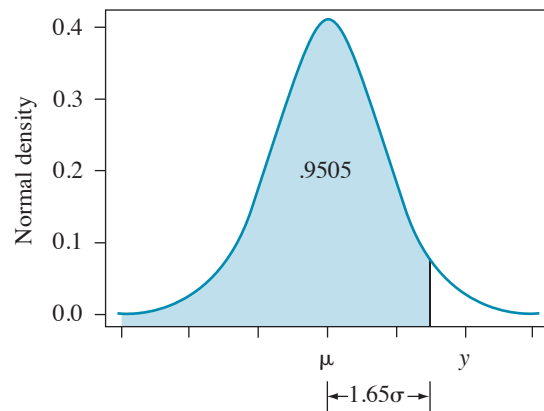
Area under a normal curve
as given in Appendix
Table 1



To find the probability that a normal random variable will lie to the left of a point 1.65 standard deviations above the mean, we look up the table entry corresponding to $z = 1.65$. This probability is .9505 (see Figure 4.11).

FIGURE 4.11

Area under a normal curve
from μ to a point 1.65
standard deviations above
the mean



To determine the probability that a measurement will be less than some value y , we first calculate the number of standard deviations that y lies away from the mean by using the formula

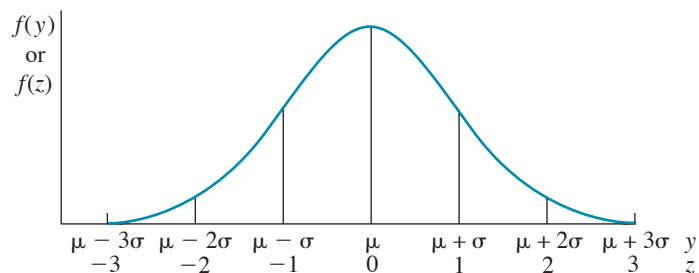
$$z = \frac{y - \mu}{\sigma}$$

z score

The value of z computed using this formula is sometimes referred to as the **z score** associated with the y -value. Using the computed value of z , we determine the appropriate probability by using Table 1 in the Appendix. Note that we are merely coding the value y by subtracting μ and dividing by σ . (In other words, $y = z\sigma + \mu$.) Figure 4.12 illustrates the values of z corresponding to specific values

FIGURE 4.12

Relationship between
specific values of y and
 $z = (y - \mu)/\sigma$



of y . Thus, a value of y that is 2 standard deviations below (to the left of) μ corresponds to $z = -2$.

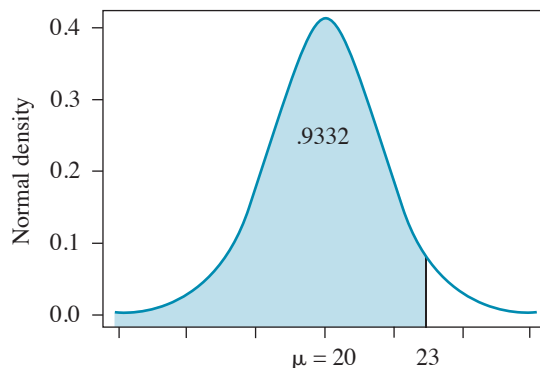
EXAMPLE 4.12

Consider a normal distribution with $\mu = 20$ and $\sigma = 2$. Determine the probability that a measurement will be less than 23.

Solution When first working problems of this type, it might be a good idea to draw a picture so that you can see the area in question, as we have in Figure 4.13.

FIGURE 4.13

Area less than $y = 23$ under normal curve, with $\mu = 20$, $\sigma = 2$



To determine the area under the curve to the left of the value $y = 23$, we first calculate the number of standard deviations $y = 23$ lies away from the mean.

$$z = \frac{y - \mu}{\sigma} = \frac{23 - 20}{2} = 1.5$$

Thus, $y = 23$ lies 1.5 standard deviations above $\mu = 20$. Referring to Table 1 in the Appendix, we find the area corresponding to $z = 1.5$ to be .9332. This is the probability that a measurement is less than 23.

EXAMPLE 4.13

For the normal distribution of Example 4.12 with $\mu = 20$ and $\sigma = 2$, find the probability that y will be less than 16.

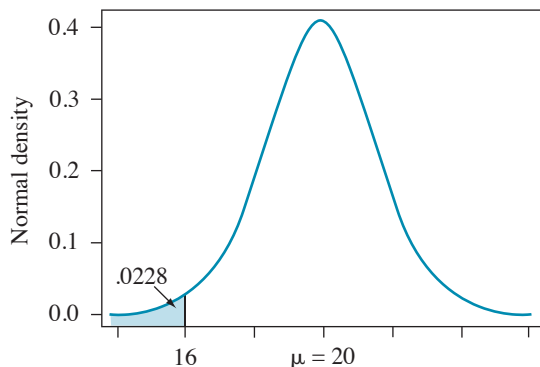
Solution In determining the area to the left of 16, we use

$$z = \frac{y - \mu}{\sigma} = \frac{16 - 20}{2} = -2$$

We find the appropriate area from Table 1 to be .0228; thus, .0228 is the probability that a measurement is less than 16. The area is shown in Figure 4.14.

FIGURE 4.14

Area less than $y = 16$ under normal curve, with $\mu = 20$, $\sigma = 2$



EXAMPLE 4.14

The mean daily milk production of a herd of Guernsey cows has a normal distribution with $\mu = 70$ pounds and $\sigma = 13$ pounds.

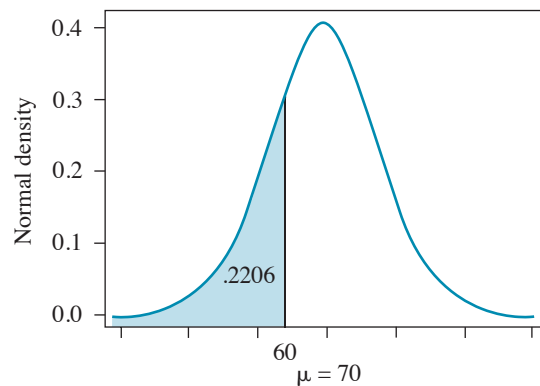
- What is the probability that the milk production for a cow chosen at random will be less than 60 pounds?
- What is the probability that the milk production for a cow chosen at random will be greater than 90 pounds?
- What is the probability that the milk production for a cow chosen at random will be between 60 pounds and 90 pounds?

Solution We begin by drawing pictures of the areas we are looking for (Figures 4.15 (a)–(c)). To answer part (a) we must compute the z values corresponding to the value of 60. The value $y = 60$ corresponds to a z score of

$$z = \frac{y - \mu}{\sigma} = \frac{60 - 70}{13} = -.77$$

From Table 1, the area to the left of 60 is .2206 (see Figure 4.15(a)).

FIGURE 4.15(a)
Area less than $y = 60$ under
normal curve, with $\mu = 70$,
 $\sigma = 13$

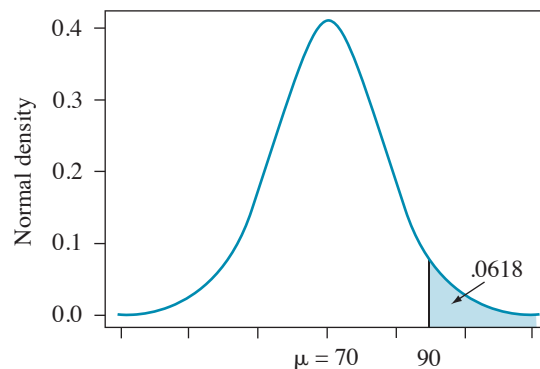


To answer part (b), the value $y = 90$ corresponds to a z score of

$$z = \frac{y - \mu}{\sigma} = \frac{90 - 70}{13} = 1.54$$

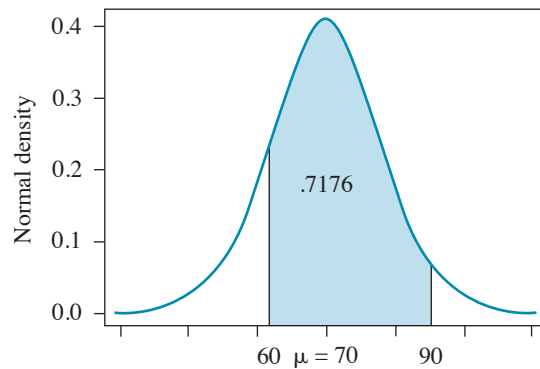
so from Table 1 we obtain .9382, the tabulated area less than 90. Thus, the area greater than 90 must be $1 - .9382 = .0618$, since the total area under the curve is 1 (see Figure 4.15(b)).

FIGURE 4.15(b)
Area greater than $y = 90$
under normal curve, with
 $\mu = 70$, $\sigma = 13$



To answer part (c), we can use our results from (a) and (b). The area between two values y_1 and y_2 is determined by finding the difference between the areas to the left of the two values, (see Figure 4.15(c)). We have the area less than 60 is .2206, and the area less than 90 is .9382. Hence, the area between 60 and 90 is $.9382 - .2206 = .7176$. We can thus conclude that 22.06% of cow production is less than 60 pounds, 6.18% is greater than 90 pounds, and 71.76% is between 60 and 90 pounds.

FIGURE 4.15(c)
Area between 60 and 90
under normal curve, with
 $\mu = 70$, $\sigma = 13$



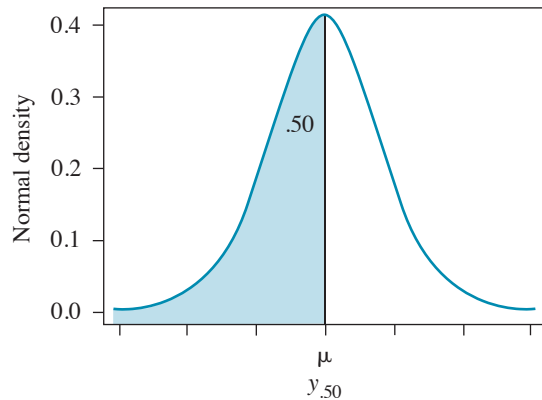
100th percentile

An important aspect of the normal distribution is that we can easily find the percentiles of the distribution. The **100th percentile** of a distribution is that value, y_p , such that 100 p % of the population values fall below y_p and 100 (1 - p) % are above y_p . For example, the median of a population is the 50th percentile, $y_{.50}$, and the quartiles are the 25th and 75th percentiles. The normal distribution is symmetric, so the median and the mean are the same value, $y_{.50} = \mu$ (see Figure 4.16(a)).

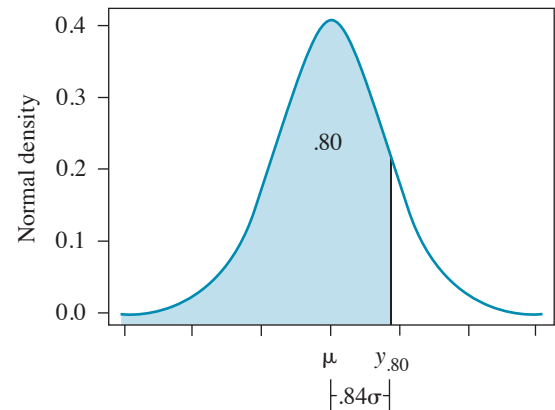
To find the percentiles of the standard normal distribution, we reverse our usage of Table 1. To find the 100 p th percentile, z_p , we find the probability p in Table 1 and then read out its corresponding number, z_p , along the margins of the table. For example, to find the 80th percentile, $z_{.80}$, we locate the probability, $p = .8000$ in Table 1. The value nearest to .8000 is .7995, which corresponds to a z value of 0.84. Thus, $z_{.80} = 0.84$ (see Figure 4.16 (b)). Now, to find the 100 p th percentile, y_p , of a normal distribution with mean μ and standard deviation σ , we need to apply the reverse of our standardization formula,

$$y_p = \mu + z_p \sigma$$

FIGURE 4.16



(a) For the normal curve, the mean and median agree



(b) The 80th percentile for the normal curve

Suppose we wanted to determine the 80th percentile of a population having a normal distribution with $\mu = 55$ and $\sigma = 3$. We have determined that $z_{.80} = 0.84$; thus, the 80th percentile for the population would be $y_{.80} = 55 + (.84)(3) = 57.52$.

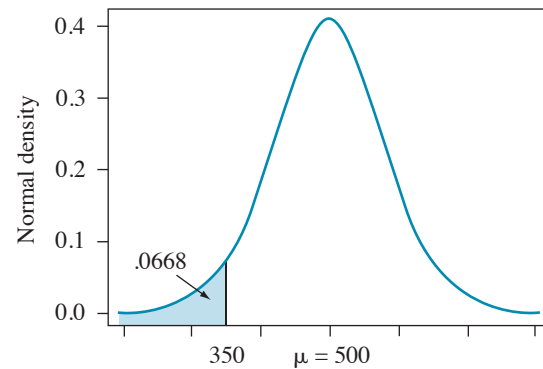
EXAMPLE 4.15

The Scholastic Assessment Test (SAT) is an exam used to measure a person's readiness for college. The mathematics scores are scaled to have a normal distribution with mean 500 and standard deviation 100. What proportion of the people taking the SAT will score below 350? To identify a group of students needing remedial assistance, we want to determine the lower 10% of all scores; that is, we want to determine the tenth percentile, $y_{.10}$.

Solution To find the proportion of scores below 350 (see Figure 4.17(a)), we need to find the area below 350:

$$z = \frac{y - \mu}{\sigma} = \frac{350 - 500}{100} = -1.5$$

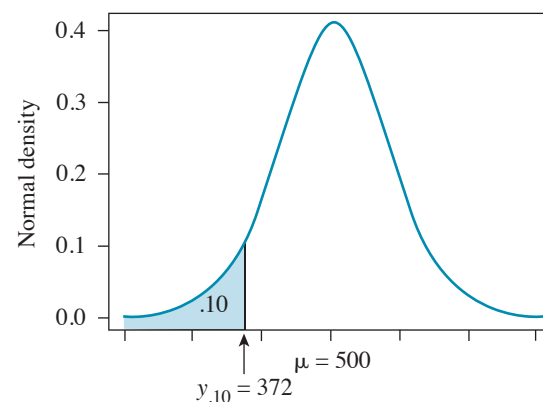
FIGURE 4.17(a)
Area less than 350 under
normal curve, with $\mu = 500$,
 $\sigma = 100$



In a normal distribution, the area to the left of a value -1.5 standard deviations from the mean is, from Table 1, .0668. Hence, 6.68%, or approximately 7%, of the persons taking the exam scored below 350. The score of 350 would be approximately the seventh percentile, $y_{.07}$, of the population of all scores.

To find the tenth percentile (see Figure 4.17(b)), we first find $z_{.10}$ in Table 1. Since .1003 is the value nearest .1000 and its corresponding z value is -1.28 ,

FIGURE 4.17(b)
The tenth percentile for a
normal curve, with $\mu = 500$,
 $\sigma = 100$



we take $z_{.10} = -1.28$. We then compute

$$y_{.10} = \mu + z_{.10}\sigma = 500 + (-1.28)(100) = 500 - 128 = 372$$

Thus, 10% of the scores on the SAT are less than 372.

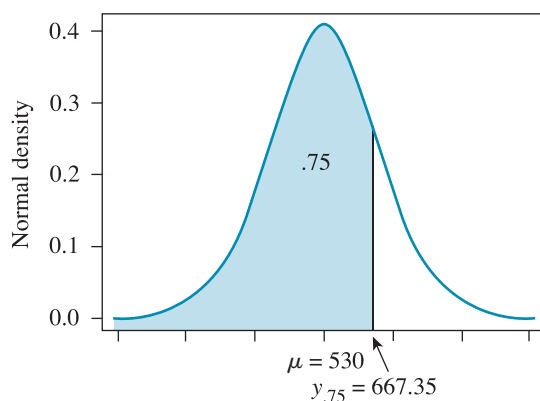
EXAMPLE 4.16

An analysis of income tax returns from the previous year indicates that for a given income classification, the amount of money owed to the government over and above the amount paid in the estimated tax vouchers for the first three payments is approximately normally distributed with a mean of \$530 and a standard deviation of \$205. Find the 75th percentile for this distribution of measurements. The government wants to target that group of returns having the largest 25% of amounts owed.

Solution We need to determine the 75th percentile, $y_{.75}$, (Figure 4.18). From Table 1, we find $z_{.75} = .67$ because the probability nearest .7500 is .7486, which corresponds to a z score of .67. We then compute

$$y_{.75} = \mu + z_{.75}\sigma = 530 + (.67)(205) = 667.35$$

FIGURE 4.18
The 75th percentile for a normal curve, with $\mu = 530$, $\sigma = 205$



Thus, 25% of the tax returns in this classification exceed \$667.35 in the amount owed the government.

EXERCISES

Basic Techniques

4.52 Use Table 1 of the Appendix to find the area under the normal curve between these values:

- a. $z = 0$ and $z = 1.3$
- b. $z = 0$ and $z = -1.9$

4.53 Repeat Exercise 4.52 for these values:

- a. $z = 0$ and $z = .7$
- b. $z = 0$ and $z = -1.2$

4.54 Repeat Exercise 4.52 for these values:

- a. $z = 0$ and $z = 1.29$
- b. $z = 0$ and $z = -.77$

4.55 Repeat Exercise 4.52 for these values:

- a. $z = -.21$ and $z = 1.35$
- b. $z = .37$ and $z = 1.20$

- 4.56** Repeat Exercise 4.52 for these values:
- $z = 1.43$ and $z = 2.01$
 - $z = -1.74$ and $z = -.75$
- 4.57** Find the probability that z is greater than 1.75.
- 4.58** Find the probability that z is less than 1.14.
- 4.59** Find a value for z , say z_0 , such that $P(z > z_0) = .5$.
- 4.60** Find a value for z , say z_0 , such that $P(z > z_0) = .025$.
- 4.61** Find a value for z , say z_0 , such that $P(z > z_0) = .0089$.
- 4.62** Find a value for z , say z_0 , such that $P(z > z_0) = .05$.
- 4.63** Find a value for z , say z_0 , such that $P(-z_0 < z < z_0) = .95$.
- 4.64** Let y be a normal random variable with mean equal to 100 and standard deviation equal to 8. Find the following probabilities:
- $P(y > 100)$
 - $P(y > 110)$
 - $P(y < 115)$
 - $P(88 < y < 120)$
 - $P(100 < y < 108)$
- 4.65** Let y be a normal random variable with $\mu = 500$ and $\sigma = 100$. Find the following probabilities:
- $P(500 < y < 696)$
 - $P(y > 696)$
 - $P(304 < y < 696)$
 - k such that $P(500 - k < y < 500 + k) = .60$
- 4.66** Suppose that y is a normal random variable with $\mu = 100$ and $\sigma = 15$.
- Show that $y < 130$ is equivalent to $z < 2$.
 - Convert $y > 82.5$ to the z -score equivalent.
 - Find $P(y < 130)$ and $P(y > 82.5)$.
 - Find $P(y > 106)$, $P(y < 94)$, and $P(94 < y < 106)$.
 - Find $P(y < 70)$, $P(y > 130)$, and $P(70 < y < 130)$.
- 4.67** Use Table 1 in the Appendix to calculate the area under the curve between these values:
- $z = 0$ and $z = 1.5$
 - $z = 0$ and $z = 1.8$
- 4.68** Repeat Exercise 4.67 for these values:
- $z = -1.96$ and $z = 1.96$
 - $z = -2.33$ and $z = 2.33$
- 4.69** What is the value of z with an area of .05 to its right? To its left? (*Hint:* Use Table 2 in the Appendix.)
- 4.70** Find the value of z for these areas.
- an area .01 to the right of z
 - an area .10 to the left of z
- 4.71** Find the probability of observing a value of z greater than these values.
- 1.96
 - 2.21
 - 2.86
 - 0.73
- 4.72** Find the probability of observing a value of z less than these values.
- 1.20
 - 2.62
 - 1.84
 - 2.17

Applications

- Gov.** **4.73** Records maintained by the office of budget in a particular state indicate that the amount of time elapsed between the submission of travel vouchers and the final reimbursement of funds has approximately a normal distribution with a mean of 39 days and a standard deviation of 6 days.
- What is the probability that the elapsed time between submission and reimbursement will exceed 50 days?
 - If you had a travel voucher submitted more than 55 days ago, what might you conclude?
- Edu.** **4.74** The College Boards, which are administered each year to many thousands of high school students, are scored so as to yield a mean of 500 and a standard deviation of 100. These scores are close to being normally distributed. What percentage of the scores can be expected to satisfy each condition?
- greater than 600
 - greater than 700
 - less than 450
 - between 450 and 600
- Bus.** **4.75** Monthly sales figures for a particular food industry tend to be normally distributed with mean of 150 (thousand dollars) and a standard deviation of 35 (thousand dollars). Compute the following probabilities:
- $P(y > 200)$
 - $P(y > 220)$
 - $P(y < 120)$
 - $P(100 < y < 200)$
- 4.76** Refer to Exercise 4.74. An exclusive club wishes to invite those scoring in the top 10% on the College Boards to join.
- What score is required to be invited to join the club?
 - What score separates the top 60% of the population from the bottom 40%? What do we call this value?
- 4.77** The mean for a normal distribution is 50 and the standard deviation is 10.
- What percentile is the value of 38? Choose the appropriate answer.
- 88.49 38.49 49.99 0.01 11.51
- Which of the following is the z score corresponding to the 67th percentile?
- 1.00 0.95 0.44 2.25 none of these
- Soc.** **4.78** The distribution of weights of a large group of high school boys is normal with $\mu = 120$ pounds and $\sigma = 10$ pounds. Which of the following is true?
- About 16% of the boys will be over 130 pounds.
 - Probably fewer than 2.5% of the boys will be below 100 pounds.
 - Half of the boys can be expected to weigh less than 120 pounds.
 - All the above are true.

4.11 Random Sampling

Thus far in the text, we have discussed random samples and introduced various sampling schemes in Chapter 2. What is the importance of random sampling? We must know how the sample was selected so we can determine probabilities associated with various sample outcomes. The probabilities of samples selected *in a*

random manner can be determined, and we can use these probabilities to make inferences about the population from which the sample was drawn.

Sample data selected in a nonrandom fashion are frequently distorted by a *selection bias*. A selection bias exists whenever there is a systematic tendency to overrepresent or underrepresent some part of the population. For example, a survey of households conducted during the week entirely between the hours of 9 A.M. and 5 P.M. would be severely biased toward households with at least one member at home. Hence, any inferences made from the sample data would be biased toward the attributes or opinions of those families with at least one member at home and may not be truly representative of the population of households in the region.

random sample

Now we turn to a definition of a **random sample** of n measurements selected from a population containing N measurements ($N > n$). (*Note:* This is a simple random sample as discussed in Chapter 2. Since most of the random samples discussed in this text will be simple random samples, we'll drop the adjective unless needed for clarification.)

DEFINITION 4.13

A sample of n measurements selected from a population is said to be a **random sample** if every different sample of size n from the population has an equal probability of being selected.

EXAMPLE 4.17

A study of crimes related to handguns is being planned for the ten largest cities in the United States. The study will randomly select two of the ten largest cities for an in-depth study following the preliminary findings. The population of interest is the ten largest cities $\{C_1, C_2, C_3, C_4, C_5, C_6, C_7, C_8, C_9, C_{10}\}$. List all possible different samples consisting of two cities that could be selected from the population of ten cities. Give the probability associated with each sample in a random sample of $n = 2$ cities selected from the population.

Solution All possible samples are listed next.

Sample	Cities	Sample	Cities	Sample	Cities
1	C_1, C_2	16	C_2, C_9	31	C_5, C_6
2	C_1, C_3	17	C_2, C_{10}	32	C_5, C_7
3	C_1, C_4	18	C_3, C_4	33	C_5, C_8
4	C_1, C_5	19	C_3, C_5	34	C_5, C_9
5	C_1, C_6	20	C_3, C_6	35	C_5, C_{10}
6	C_1, C_7	21	C_3, C_7	36	C_6, C_7
7	C_1, C_8	22	C_3, C_8	37	C_6, C_8
8	C_1, C_9	23	C_3, C_9	38	C_6, C_9
9	C_1, C_{10}	24	C_3, C_{10}	39	C_6, C_{10}
10	C_2, C_3	25	C_4, C_5	40	C_7, C_8
11	C_2, C_4	26	C_4, C_6	41	C_7, C_9
12	C_2, C_5	27	C_4, C_7	42	C_7, C_{10}
13	C_2, C_6	28	C_4, C_8	43	C_8, C_9
14	C_2, C_7	29	C_4, C_9	44	C_8, C_{10}
15	C_2, C_8	30	C_4, C_{10}	45	C_9, C_{10}

random number table

Now, let us suppose that we select a random sample of $n = 2$ cities from the 45 possible samples. The sample selected is called a *random* sample if every sample has an equal probability, $1/45$, of being selected.

One of the simplest and most reliable ways to select a random sample of n measurements from a population is to use a table of random numbers (see Table 13 in the Appendix). **Random number tables** are constructed in such a way that, no matter where you start in the table and no matter in which direction you move, the digits occur randomly and with equal probability. Thus, if we wished to choose a random sample of $n = 10$ measurements from a population containing 100 measurements, we could label the measurements in the population from 0 to 99 (or 1 to 100). Then by referring to Table 13 in the Appendix and choosing a random starting point, the next ten two-digit numbers going across the page would indicate the labels of the particular measurements to be included in the random sample. Similarly, by moving up or down the page, we would also obtain a random sample.

This listing of all possible samples is feasible only when both the sample size n and the population size N are small. We can determine the number, M , of distinct samples of size n that can be selected from a population of N measurements using the following formula:

$$M = \frac{N!}{n!(N-n)!}$$

In Example 4.17, we had $N = 10$ and $n = 2$. Thus,

$$M = \frac{10!}{2!(10-2)!} = \frac{10!}{2!8!} = 45$$

The value of M becomes very large even when N is fairly small. For example, if $N = 50$ and $n = 5$, then $M = 2,118,760$. Thus, it would be very impractical to list all 2,118,760 possible samples consisting of $n = 5$ measurements from a population of $N = 50$ measurements and then randomly select one of the samples. In practice, we construct a list of elements in the population by assigning a number from 1 to N to each element in the population, called the *sampling frame*. We then randomly select n integers from the integers $(1, 2, \dots, N)$ by using a table of random numbers (see Table 13 in the Appendix) or by using a computer program. Most statistical software programs contain routines for randomly selecting n integers from the integers $(1, 2, \dots, N)$, where $N > n$. Exercise 4.86 contains the necessary commands for using Minitab to generate the random sample.

EXAMPLE 4.18

A small community consists of 850 families. We wish to obtain a random sample of 20 families to ascertain public acceptance of a wage and price freeze. Refer to Table 13 in the Appendix to determine which families should be sampled.

Solution Assuming that a list of all families in the community is available (such as a telephone directory), we can label the families from 0 to 849 (or, equivalently, from 1 to 850). Then, referring to Table 13 in the Appendix, we choose a starting point. Suppose we have decided to start at line 1, column 3. Going down the page we choose the first 20 three-digit numbers between 000 and 849. From Table 13, we have

015	110	482	333
255	564	526	463
225	054	710	337
062	636	518	224
818	533	524	055

These 20 numbers identify the 20 families that are to be included in our sample.

A telephone directory is not always the best source for names, especially in surveys related to economics or politics. In the 1936 presidential campaign, Franklin Roosevelt was running as the Democratic candidate against the Republican candidate, Governor Alfred Landon of Kansas. This was a difficult time for the nation; the country had not yet recovered from the Great Depression of the 1930s, and there were still 9 million people unemployed.

The *Literary Digest* set out to sample the voting public and predict the winner of the election. Using names and addresses taken from telephone books and club memberships, the *Literary Digest* sent out 10 million questionnaires and got 2.4 million back. Based on the responses to the questionnaire, the *Digest* predicted a Landon victory by 57% to 43%.

At this time, George Gallup was starting his survey business. He conducted two surveys. The first one, based on 3,000 people, predicted what the results of the *Digest* survey would be long before the *Digest* results were published; the second survey, based on 50,000, was used to forecast *correctly* the Roosevelt victory.

How did Gallup correctly predict what the *Literary Digest* survey would predict and then, with another survey, correctly predict the outcome of the election? Where did the *Literary Digest* go wrong? The first problem was a severe selection bias. By taking the names and addresses from telephone directories and club memberships, its survey systematically excluded the poor. Unfortunately for the *Digest*, the vote was split along economic lines; the poor gave Roosevelt a large majority, whereas the rich tended to vote for Landon. A second reason for the error could be due to a *nonresponse bias*. Because only 20% of the 10 million people returned their surveys, and approximately half of those responding favored Landon, one might suspect that maybe the nonrespondents had different preferences than did the respondents. This was, in fact, true.

How, then does one achieve a random sample? Careful planning and a certain amount of ingenuity are required to have even a decent chance to approximate random sampling. This is especially true when the universe of interest involves people. People can be difficult to work with; they have a tendency to discard mail questionnaires and refuse to participate in personal interviews. Unless we are very careful, the data we obtain may be full of biases having unknown effects on the inferences we are attempting to make.

We do not have sufficient time to explore the topic of random sampling further in this text; entire courses at the undergraduate and graduate levels can be devoted to sample survey research methodology. The important point to remember is that data from a random sample will provide the foundation for making statistical inferences in later chapters. Random samples are not easy to obtain, but with care we can avoid many potential biases that could affect the inferences we make.

EXERCISES Basic Techniques

4.79 Define what is meant by a random sample. Is it possible to draw a truly random sample? Comment.

Gov. 4.80 Suppose that we want to select a random sample of $n = 10$ persons from a population of 800. Use Table 13 in the Appendix to identify the persons to appear in the sample.

Pol. Sci. 4.81 Refer to Exercise 4.80. Identify the elements of a population of $N = 1,000$ to be included in a random sample of $n = 15$.

Applications

Soc. 4.82 City officials want to sample the opinions of the homeowners in a community regarding the desirability of increasing local taxes to improve the quality of the public schools. If a random number table is used to identify the homes to be sampled and a home is discarded if the homeowner is not home when visited by the interviewer, is it likely this process will approximate random sampling? Explain.

4.83 A local TV network wants to run an informal survey of individuals who exit from a local voting station to ascertain early results on a proposal to raise funds to move the city-owned historical museum to a new location. How might the network sample voters to approximate random sampling?

4.84 A psychologist was interested in studying women who are in the process of obtaining a divorce to determine whether the women experienced significant attitudinal changes after the divorce has been finalized. Existing records from the geographic area in question show that 798 couples have recently filed for divorce. Assume that a sample of 25 women is needed for the study, and use Table 13 in the Appendix to determine which women should be asked to participate in the study. (*Hint:* Begin in column 2, row 1, and proceed down.)

4.85 Refer to Exercise 4.84. As is the case in most surveys, not all persons chosen for a study will agree to participate. Suppose that 5 of the 25 women selected refuse to participate. Determine 5 more women to be included in the study.

4.86 Suppose you have been asked to run a public opinion poll related to an upcoming election. There are 230 precincts in the city, and you need to randomly select 50 registered voters from each precinct. Suppose that each precinct has 1,000 registered voters and it is possible to obtain a list of these persons. You assign the numbers 1 to 1,000 to the 1,000 people on each list, with 1 to the first person on the list and 1,000 to the last person. You need to next obtain a random sample of 50 numbers from the numbers 1 to 1,000. The names on the sampling frame corresponding to these 50 numbers will be the 50 persons selected for the poll. A Minitab program is shown here for purposes of illustration. Note that you would need to run this program 230 separate times to obtain a new random sample for each of the 230 precincts.

Follow these steps:

Click on **Calc**.

Click on **Random Data**.

Click on **Integer**.

Type 5 in the **Generate rows of data** box.

Type **c1–c10** in the **Store in Column(s)** box.

Type **1** in the **Minimum value** box.

Type **1000** in the **Maximum value** box.

Click on **OK**.

Click on **File**.

Click on **Print Worksheet**.

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
1	340	701	684	393	313	312	834	596	321	739
2	783	877	724	498	315	282	175	611	725	571
3	862	625	971	30	766	256	40	158	444	546
4	974	402	768	593	980	536	483	244	51	201
5	232	742	1	861	335	129	409	724	340	218

- Using either a random number table or a computer program, generate a second random sample of 50 numbers from the numbers 1 to 1,000.
- Give several reasons why you need to generate a different set of random numbers for each of the precincts. Why not use the same set of 50 numbers for all 230 precincts?

4.12 Sampling Distributions

We discussed several different measures of central tendency and variability in Chapter 3 and distinguished between numerical descriptive measures of a population (parameters) and numerical descriptive measures of a sample (statistics). Thus, μ and σ are parameters, whereas $\bar{y}$ and s are statistics.

The numerical value of a sample statistic cannot be predicted exactly in advance. Even if we knew that a population mean μ was \$216.37 and that the population standard deviation σ was \$32.90—even if we knew the complete population distribution—we could not say that the sample mean $\bar{y}$ would be exactly equal to \$216.37. A sample statistic is a random variable; it is subject to random variation because it is based on a random sample of measurements selected from the population of interest. Also, like any other random variable, a sample statistic has a probability distribution. We call the probability distribution of a sample statistic the *sampling distribution* of that statistic. Stated differently, the sampling distribution of a statistic is the population of all possible values for that statistic.

The actual mathematical derivation of sampling distributions is one of the basic problems of mathematical statistics. We will illustrate how the sampling distribution for $\bar{y}$ can be obtained for a simplified population. Later in the chapter, we will present several general results.

EXAMPLE 4.19

The sample $\bar{y}$ is to be calculated from a random sample of size 2 taken from a population consisting of ten values (2, 3, 4, 5, 6, 7, 8, 9, 10, 11). Find the sampling distribution of $\bar{y}$, based on a random sample of size 2.

Solution One way to find the sampling distribution is by counting. There are 45 possible samples of two items selected from the ten items. These are shown here:

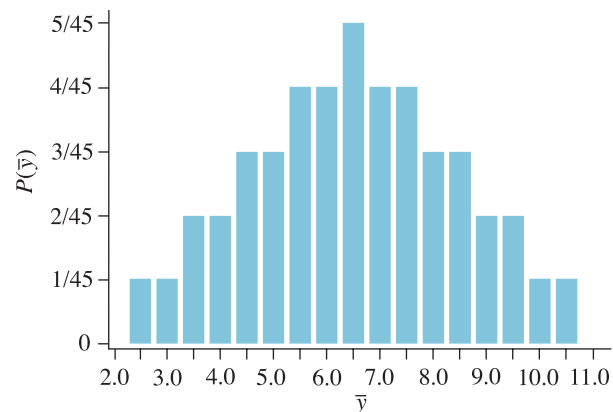
Sample	Value of $\bar{y}$	Sample	Value of $\bar{y}$	Sample	Value of $\bar{y}$
2, 3	2.5	3, 10	6.5	6, 7	6.5
2, 4	3	3, 11	7	6, 8	7
2, 5	3.5	4, 5	4.5	6, 9	7.5
2, 6	4	4, 6	5	6, 10	8
2, 7	4.5	4, 7	5.5	6, 11	8.5
2, 8	5	4, 8	6	7, 8	7.5
2, 9	5.5	4, 9	6.5	7, 9	8
2, 10	6	4, 10	7	7, 10	8.5
2, 11	6.5	4, 11	7.5	7, 11	9
3, 4	3.5	5, 6	5.5	8, 9	8.5
3, 5	4	5, 7	6	8, 10	9
3, 6	4.5	5, 8	6.5	8, 11	9.5
3, 7	5	5, 9	7	9, 10	9.5
3, 8	5.5	5, 10	7.5	9, 11	10
3, 9	6	5, 11	8	10, 11	10.5

Assuming each sample of size 2 is equally likely, it follows that the sampling distribution for $\bar{y}$ based on $n = 2$ observations selected from the population $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$ is as indicated here.

$\bar{y}$	$P(\bar{y})$	$\bar{y}$	$P(\bar{y})$
2.5	1/45	7	4/45
3	1/45	7.5	4/45
3.5	2/45	8	3/45
4	2/45	8.5	3/45
4.5	3/45	9	2/45
5	3/45	9.5	2/45
5.5	4/45	10	1/45
6	4/45	10.5	1/45
6.5	5/45		

The sampling distribution is shown as a graph in Figure 4.19. Note that the distribution is symmetric, with a mean of 6.5 and a standard deviation of approximately 2.0 (the range divided by 4).

FIGURE 4.19
Sampling distribution for $\bar{y}$



Example 4.19 illustrates for a very small population that we could in fact enumerate every possible sample of size 2 selected from the population and then compute all possible values of the sample mean. The next example will illustrate the properties of the sample mean, $\bar{y}$, when sampling from a larger population. This example will illustrate that the behavior of $\bar{y}$ as an estimator of μ depends on the sample size, n . Later in this chapter, we will illustrate the effect of the shape of the population distribution on the sampling distribution of $\bar{y}$.

EXAMPLE 4.20

In this example, the population values are known and, hence, we can compute the exact values of the population mean, μ , and population standard deviation, σ . We will then examine the behavior of $\bar{y}$ based on samples of size $n = 5$, 10, and 25 selected from the population. The population consists of 500 pennies from which we compute the age of each penny: Age = 2000 – Date on penny. The histogram of the 500 ages is displayed in Figure 4.20(a). The shape is skewed to the right with a very long right tail. The mean and standard deviation are computed to be $\mu = 13.468$ years and $\sigma = 11.164$ years. In order to generate the sampling distribution of $\bar{y}$ for $n = 5$, we would need to generate all possible samples of size $n = 5$ and then compute the $\bar{y}$ from each of these samples. This would be an enormous task since there are 255,244,687,600 possible samples of size 5 that could be selected from a population of 500 elements. The number of possible samples of size 10 or 25 is so large it makes even the national debt look small. Thus, we will use a computer program, *S-plus*, to select 25,000 samples of size 5 from the population of 500 pennies. For example, the first sample consists of pennies with ages 4, 12, 26, 16, and 9. The sample mean $\bar{y} = (4 + 12 + 26 + 16 + 9)/5 = 13.4$. We repeat 25,000 times the process of selecting 5 pennies, recording their ages, y_1, y_2, y_3, y_4, y_5 , and then computing $\bar{y} = (y_1 + y_2 + y_3 + y_4 + y_5)/5$. The 25,000 values for $\bar{y}$ are then plotted in a frequency histogram, called the *sampling distribution* of $\bar{y}$ for $n = 5$. A similar procedure is followed for samples of size $n = 10$ and 25. The sampling distributions obtained are displayed in Figures 4.20(b)–(d).

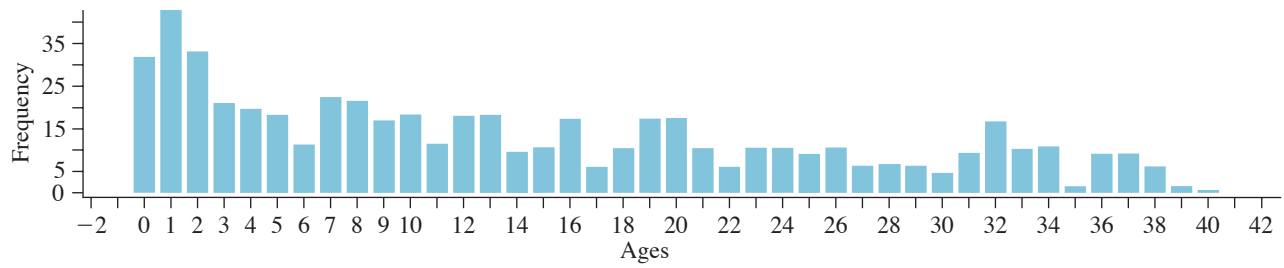
Note that all three sampling distributions have nearly the same central value, approximately 13.5. (See Table 4.5.) The mean values of $\bar{y}$ for the three samples are nearly the same as the population mean, $\mu = 13.468$. In fact, if we had generated all possible samples for all three values of n , the mean of the possible values of $\bar{y}$ would agree exactly with μ .

The next characteristic to notice about the three histograms is their shape. All three are somewhat symmetric in shape, achieving a nearly normal distribution shape when $n = 25$. However, the histogram for $\bar{y}$ based on samples of size $n = 5$ is more spread out than the histogram based on $n = 10$, which, in turn, is more spread out than the histogram based on $n = 25$. When n is small, we are much more likely to obtain a value of $\bar{y}$ far from μ than when n is larger. What causes this increased dispersion in the values of $\bar{y}$? A single extreme y , either large or small relative to μ , in the sample has a greater influence on the size of $\bar{y}$ when n is small than when n is large. Thus, sample means based on small n are less accurate in their estimation of μ than their large-sample counterparts.

Table 4.5 contains summary statistics for the sampling distribution of $\bar{y}$. The sampling distribution of $\bar{y}$ has mean $\mu_{\bar{y}}$ and standard deviation $\sigma_{\bar{y}}$, which are related to the population mean, μ , and standard deviation, σ , by the following relationship:

$$\mu_{\bar{y}} = \mu \quad \sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$$

FIGURE 4.20



(a) Histogram of ages for 500 pennies

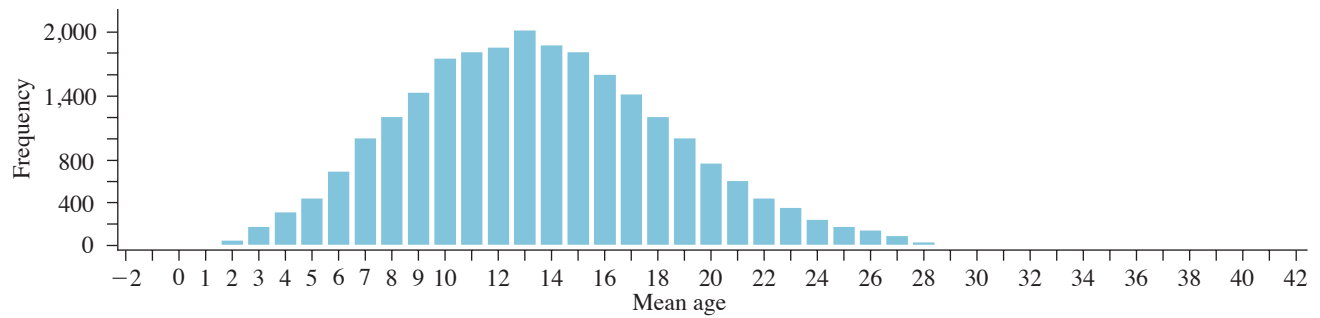
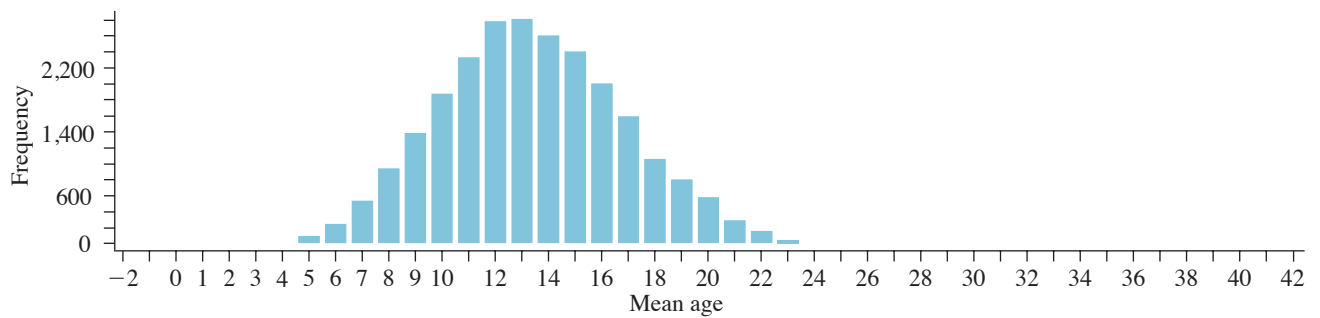
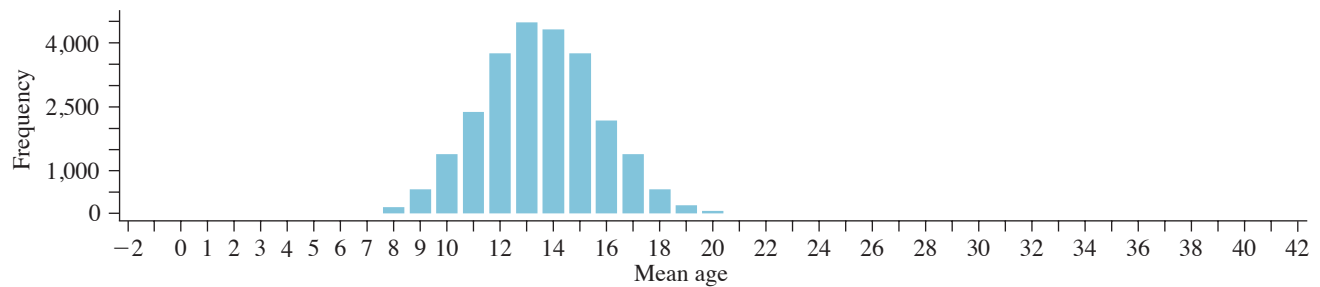
(b) Sampling distribution of $\bar{y}$ for $n = 5$ (c) Sampling distribution of $\bar{y}$ for $n = 10$ (d) Sampling distribution of $\bar{y}$ for $n = 25$

TABLE 4.5

Means and standard deviations for the sampling distributions of $\bar{y}$

Sample Size	Mean of $\bar{y}$	Standard Deviation of $\bar{y}$	$\frac{11.1638}{\sqrt{n}}$
1 (Population)	13.468 (μ)	11.1638 (σ)	11.1638
5	13.485	4.9608	4.9926
10	13.438	3.4926	3.5303
25	13.473	2.1766	2.2328

standard error of $\bar{y}$

From Table 4.5, we note that the three sampling deviations have means that are approximately equal to the population mean. Also, the three sampling deviations have standard deviations that are approximately equal to $\sigma/\sqrt{n}$. If we had generated all possible values of $\bar{y}$, then the standard deviation of $\bar{y}$ would equal $\sigma/\sqrt{n}$ exactly. This quantity, $\sigma_{\bar{y}} = \sigma/\sqrt{n}$, is called the **standard error of $\bar{y}$** .

Central Limit Theorems

Quite a few of the more common sample statistics, such as the sample median and the sample standard deviation, have sampling distributions that are nearly normal for moderately sized values of n . We can observe this behavior by computing the sample median and sample standard deviation from each of the three sets of 25,000 sample ($n = 5, 10, 25$) selected from the population of 500 pennies. The resulting sampling distributions are displayed in Figures 4.21(a)–(d), for the sample median, and Figures 4.22(a)–(d), for the sample standard deviation. The sampling distribution of both the median and the standard deviation are more highly skewed in comparison to the sampling distribution of the sample mean. In fact, the value of n at which the sampling distributions of the sample median and standard deviation have a nearly normal shape is much larger than the value required for the sample mean. A series of theorems in mathematical statistics called the **Central Limit Theorems** provide theoretical justification for our approximating the true sampling distribution of many sample statistics with the normal distribution. We will discuss one such theorem for the sample mean. Similar theorems exist for the sample median, sample standard deviation, and the sample proportion.

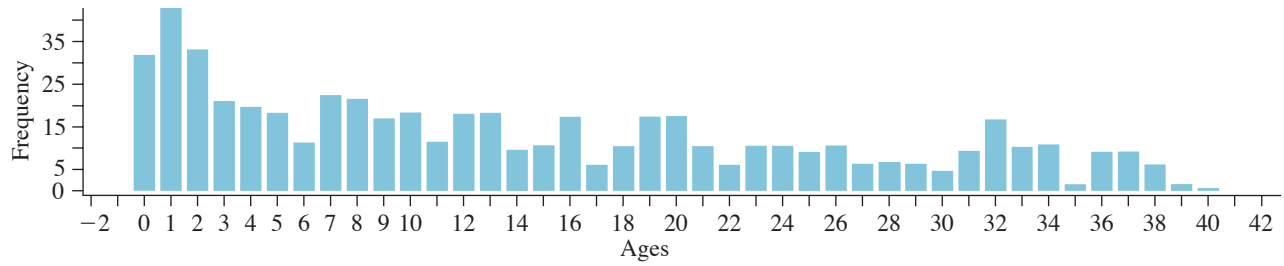
THEOREM 4.1

Central Limit Theorem for $\bar{y}$

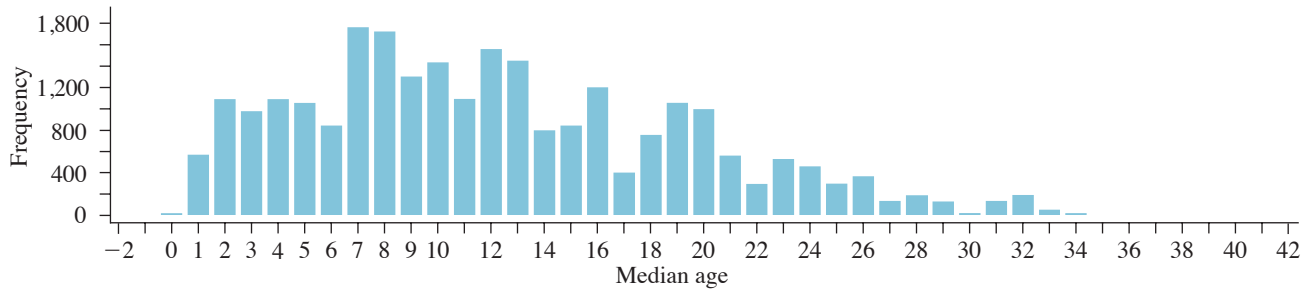
Let $\bar{y}$ denote the sample mean computed from a random sample of n measurements from a population having a mean, μ , and finite standard deviation, σ . Let $\mu_{\bar{y}}$ and $\sigma_{\bar{y}}$ denote the mean and standard deviation of the sampling distribution of $\bar{y}$, respectively. Based on repeated random samples of size n from the population, we can conclude the following:

1. $\mu_{\bar{y}} = \mu$
2. $\sigma_{\bar{y}} = \sigma/\sqrt{n}$
3. When n is large, the sampling distribution of $\bar{y}$ will be approximately normal (with the approximation becoming more precise as n increases).
4. When the population distribution is normal, the sampling distribution of $\bar{y}$ is exactly normal for any sample size n .

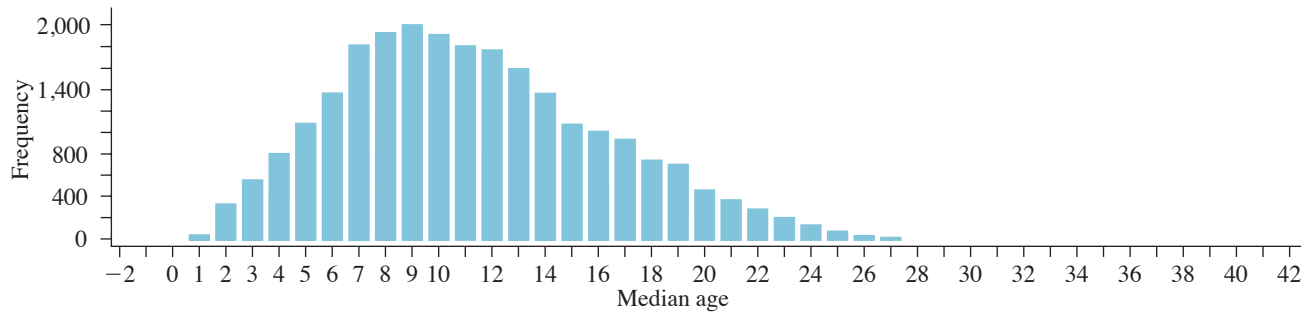
FIGURE 4.21



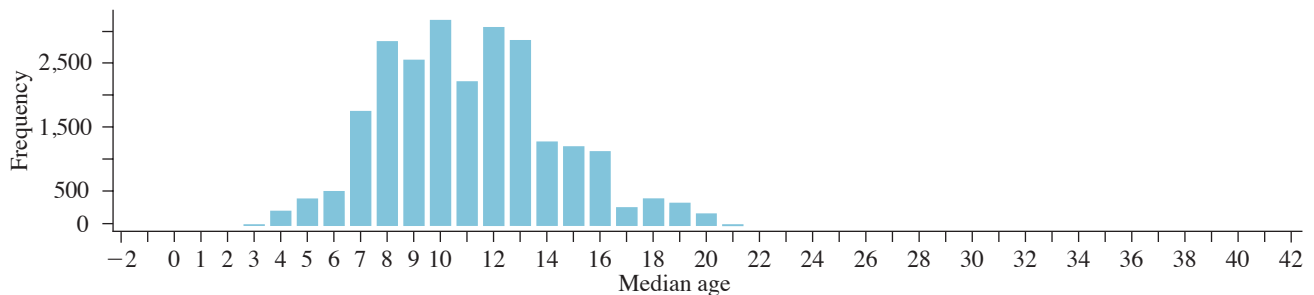
(a) Histogram of ages for 500 pennies



(b) Sampling distribution of median for $n = 5$

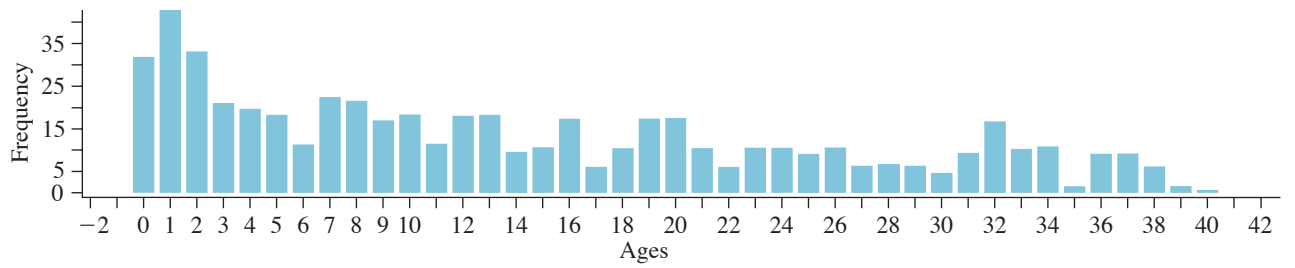


(c) Sampling distribution of median for $n = 10$



(d) Sampling distribution of median for $n = 25$

FIGURE 4.22



(a) Histogram of ages for 500 pennies

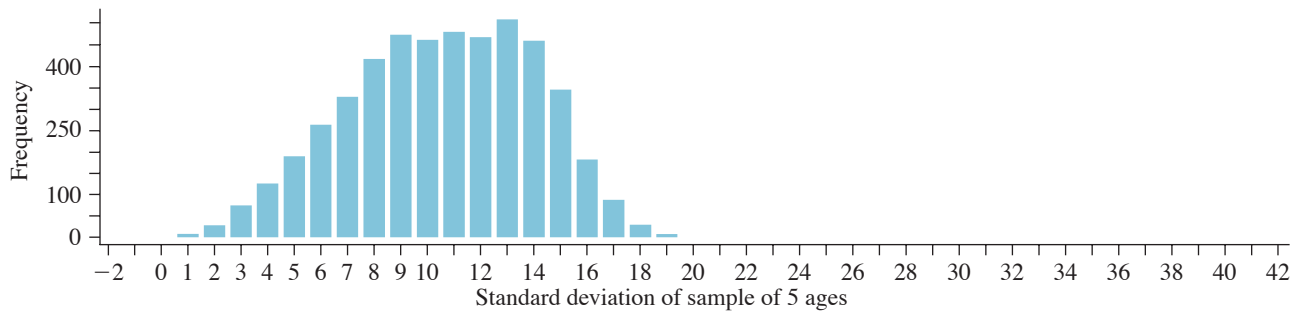
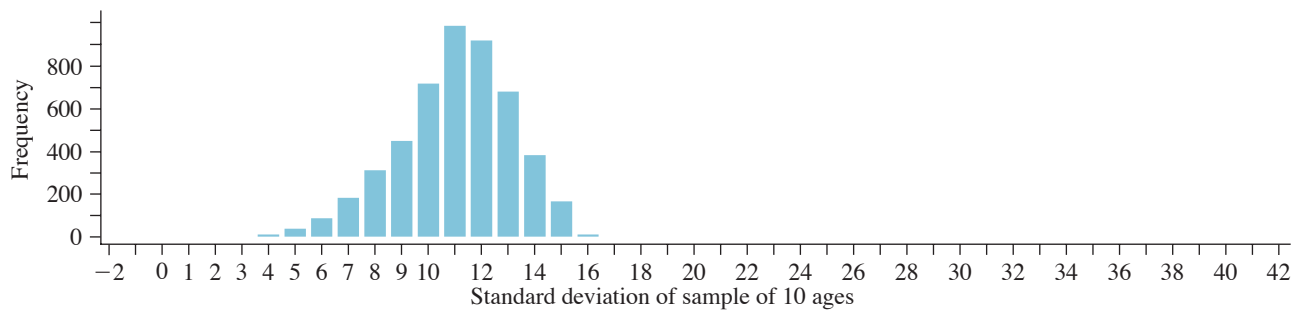
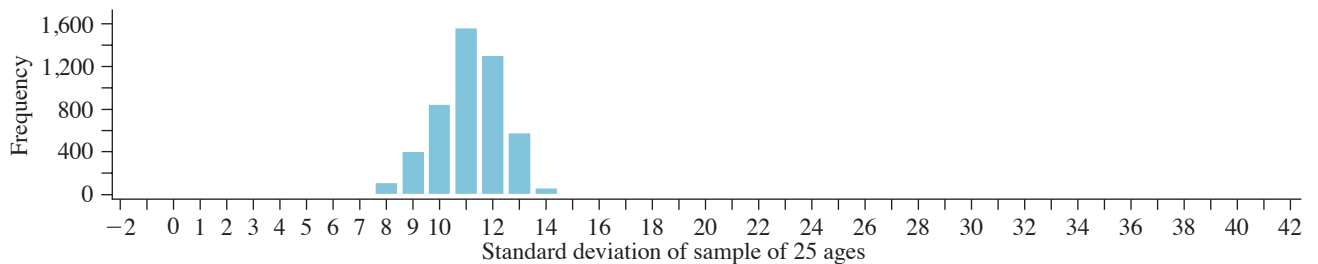
(b) Sampling distribution of standard deviation for $n = 5$ (c) Sampling distribution of standard deviation for $n = 10$ (d) Sampling distribution of standard deviation for $n = 25$

Figure 4.20 illustrates the Central Limit Theorem. Figure 4.20(a) displays the distribution of the measurements y in the population from which the samples are to be drawn. No specific shape was required for these measurements for the Central Limit Theorem to be validated. Figures 4.20(b)–(d) illustrate the sampling distribution for the sample mean $\bar{y}$ when n is 5, 10, and 25, respectively. We note that even for a very small sample size, $n = 10$, the shape of the sampling distribution of $\bar{y}$ is very similar to that of a normal distribution. This is not true in general. If the population distribution had many extreme values or several modes, the sampling distribution of $\bar{y}$ would require n to be considerably larger in order to achieve a symmetric bell shape.

We have seen that the sample size n has an effect on the shape of the sampling distribution of $\bar{y}$. The shape of the distribution of the population measurements also will affect the shape of the sampling distribution of $\bar{y}$. Figures 4.23 and 4.24 illustrate the effect of the population shape on the shape of the sampling distribution of $\bar{y}$. In Figure 4.23, the population measurements have a normal distribution. The sampling distribution of $\bar{y}$ is *exactly* a normal distribution for all values of n , as is illustrated for $n = 5, 10$, and 25 in Figure 4.23. When the population distribution is nonnormal, as depicted in Figure 4.24, the sampling distribution of $\bar{y}$ will not have a normal shape for small n (see Figure 4.24 with $n = 5$). However, for $n = 10$ and 25, the sampling distributions are nearly normal in shape, as can be seen in Figure 4.24.

FIGURE 4.23
Sampling distribution of $\bar{y}$
for $n = 5, 10, 25$ when
sampling from a normal
distribution

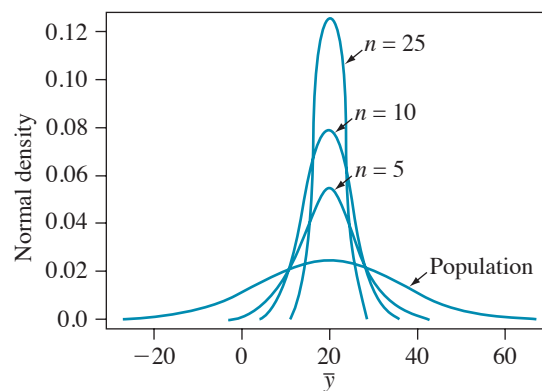
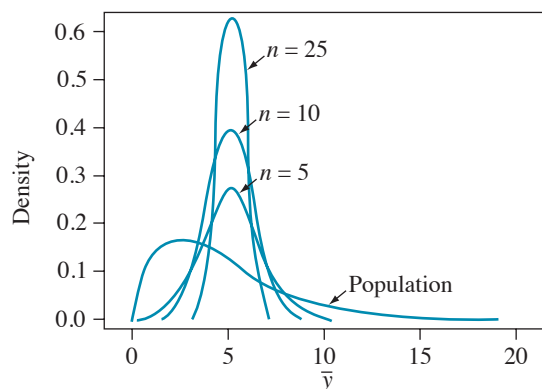


FIGURE 4.24
Sampling distribution of $\bar{y}$
for $n = 5, 10, 25$ when
sampling from a skewed
distribution



It is very unlikely that the exact shape of the population distribution will be known. Thus, the exact shape of the sampling distribution of $\bar{y}$ will not be

known either. The important point to remember is that the sampling distribution of $\bar{y}$ will be approximately normally distributed with a mean $\mu_{\bar{y}} = \mu$, the population mean, and a standard deviation $\sigma_{\bar{y}} = \sigma/\sqrt{n}$. The approximation will be more precise as n , the sample size for each sample, increases and as the shape of the population distribution becomes more like the shape of a normal distribution.

An obvious question is: How large should the sample size be for the Central Limit Theorem to hold? Numerous simulation studies have been conducted over the years and the results of these studies suggest that, in general, the Central Limit Theorem holds for $n > 30$. However, one should not apply this rule blindly. If the population is heavily skewed, the sampling distribution for $\bar{y}$ will still be skewed even for $n > 30$. On the other hand, if the population is symmetric, the Central Limit Theorem holds for $n < 30$.

Therefore, take a look at the data. If the sample histogram is clearly skewed, then the population will also probably be skewed. Consequently, a value of n much higher than 30 may be required to have the sampling distribution of $\bar{y}$ be approximately normal. Any inference based on the normality of $\bar{y}$ for $n = 30$ under this condition should be examined carefully.

As demonstrated in Figures 4.21 and 4.22, the Central Limit Theorem can be extended to many different sample statistics. The form of the Central Limit Theorem for the sample median and sample standard deviation is somewhat more complex than for the sample mean. Many of the statistics that we will encounter in later chapters will be either averages or sums of variables. The Central Limit Theorem for sums can be easily obtained from the Central Limit Theorem for the sample mean. Suppose we have a random sample of n measurements, $y_1, \dots, y_n$, from a population and let $\Sigma y = y_1 + \dots + y_n$.

THEOREM 4.2

Central Limit Theorem for Σy

Let Σy denote the sum of a random sample of n measurements from a population having a mean μ and finite standard deviation σ . Let $\mu_{\Sigma y}$ and $\sigma_{\Sigma y}$ denote the mean and standard deviation of the sampling distribution of Σy , respectively. Based on repeated random samples of size n from the population, we can conclude the following:

1. $\mu_{\Sigma y} = n\mu$
2. $\sigma_{\Sigma y} = \sqrt{n}\sigma$
3. When n is large, the sampling distribution of Σy will be approximately normal (with the approximation becoming more precise as n increases).
4. When the population distribution is normal, the sampling distribution of Σy is exactly normal for any sample size n .

Usually, a sample statistic is used as an estimate of a population parameter. For example, a sample mean $\bar{y}$ can be used to estimate the population mean μ from which the sample was selected. Similarly, a sample median and sample standard deviation estimate the corresponding population median and standard deviation. The sampling distribution of a sample statistic is then used to determine how accurate the estimate is likely to be. In Example 4.19, the population mean μ is known to be 6.5. Obviously, we do not know μ in any practical study or

experiment. However, we can use the sampling distribution of $\bar{y}$ to determine the probability that the value of $\bar{y}$ for a random sample of $n = 2$ measurements from the population will be more than three units from μ . Using the data in Example 4.19, this probability is

$$P(2.5) + P(3) + P(10) + P(10.5) = \frac{4}{45}$$

In general, we would use the normal approximation from the Central Limit Theorem in making this calculation because the sampling distribution of a sample statistic is seldom known. This type of calculation will be developed in Chapter 5. Since a sample statistic is used to make inferences about a population parameter, the sampling distribution of the statistic is crucial in determining the accuracy of the inference.

interpretations of a sampling distribution

Sampling distributions can be **interpreted** in at least two ways. One way uses the long-run relative frequency approach. Imagine taking repeated samples of a fixed size from a given population and calculating the value of the sample statistic for each sample. In the long run, the relative frequencies for the possible values of the sample statistic will approach the corresponding sampling distribution probabilities. For example, if one took a large number of samples from the population distribution corresponding to the probabilities of Example 4.19 and, for each sample, computed the sample mean, approximately 9% would have $\bar{y} = 5.5$.

The other way to interpret a sampling distribution makes use of the classical interpretation of probability. Imagine listing all possible samples that could be drawn from a given population. The probability that a sample statistic will have a particular value (say, that $\bar{y} = 5.5$) is then the proportion of all possible samples that yield that value. In Example 4.19, $P(\bar{y} = 5.5) = 4/45$ corresponds to the fact that 4 of the 45 samples have a sample mean equal to 5.5. Both the repeated-sampling and the classical approach to finding probabilities for a sample statistic are legitimate.

In practice, though, a sample is taken only once, and only one value of the sample statistic is calculated. A sampling distribution is not something you can see in practice; it is not an empirically observed distribution. Rather, it is a theoretical concept, a set of probabilities derived from assumptions about the population and about the sampling method.

There's an unfortunate similarity between the phrase "sampling distribution," meaning the theoretically derived probability distribution of a statistic, and the phrase "sample distribution," which refers to the histogram of individual values actually observed in a particular sample. The two phrases mean very different things. To avoid confusion, we will refer to the distribution of sample values as the **sample histogram** rather than as the sample distribution.

sample histogram

EXERCISES

Basic Techniques

4.87 A random sample of 16 measurements is drawn from a population with a mean of 60 and a standard deviation of 5. Describe the sampling distribution of $\bar{y}$, the sample mean. Within what interval would you expect $\bar{y}$ to lie approximately 95% of the time?

4.88 Refer to Exercise 4.87. Describe the sampling distribution for the sample sum $\sum y_i$. Is it unlikely (improbable) that $\sum y_i$ would be more than 70 units away from 960? Explain.

4.89 In Exercise 4.87, a random sample of 16 observations was to be selected from a population with $\mu = 60$ and $\sigma = 5$. Assume the original population of measurements is normally distributed. Use a computer program to simulate the sampling distribution of $\bar{y}$

based on 500 samples consisting of 16 observations each. A Minitab program that can be used in conducting the simulation is given in the last section of this chapter.

Applications

- Psy.** **4.90** Psychomotor retardation scores for a large group of manic-depressive patients were approximately normal, with a mean of 930 and a standard deviation of 130.
- What fraction of the patients scored between 800 and 1,100?
 - Less than 800?
 - Greater than 1,200?
- 4.91** Refer to Exercise 4.90.
- Find the 90th percentile for the distribution of manic-depressive scores. (*Hint:* Solve for y in the expression $z = (y - \mu)/\sigma$, where z is the number of standard deviations the 90th percentile lies above the mean μ .)
 - Find the interquartile range.
- Soc.** **4.92** Federal resources have been tentatively approved for the construction of an outpatient clinic. In order to design a facility that will handle patient load requirements and stay within a limited budget, the designers studied patient demand. From studying a similar facility in the area, they found that the distribution of the number of patients requiring hospitalization during a week could be approximated by a normal distribution with a mean of 125 and a standard deviation of 32.
- Use the Empirical Rule to describe the distribution of y , the number of patients requesting service in a week.
 - If the facility was built with a 160-patient capacity, what fraction of the weeks might the clinic be unable to handle the demand?
- 4.93** Refer to Exercise 4.92. What size facility should be built so the probability of the patient load's exceeding the clinic capacity is .05? .01?
- Soc.** **4.94** Based on the 1990 census, the number of hours per day adults spend watching television is approximately normally distributed with a mean of 5 hours and a standard deviation of 1.3 hours.
- What proportion of the population spends more than 7 hours per day watching television?
 - In a 1998 study of television viewing, a random sample of 500 adults reported that the average number of hours spent viewing television was greater than 5.5 hours per day. Do the results of this survey appear to be consistent with the 1990 census? (*Hint:* If the census results are still correct, what is the probability that the average viewing time would exceed 5.5 hours?)
- Env.** **4.95** The level of a particular pollutant, nitrogen oxide, in the exhaust of a hypothetical model of car, the Polluter, when driven in city traffic has approximately a normal distribution with a mean level of 2.1 grams per mile (g/m) and a standard deviation of 0.3 g/m.
- If the EPA mandates that a nitrogen oxide level of 2.7 g/m cannot be exceeded, what proportion of Polluters would be in violation of the mandate?
 - At most, 25% of Polluters exceed what nitrogen oxide level value (that is, find the 75th percentile)?
 - The company producing the Polluter must reduce the nitrogen oxide level so that at most 5% of its cars exceed the EPA level of 2.7 g/m. If the standard deviation remains 0.3 g/m, to what value must the mean level be reduced so that at most 5% of Polluters would exceed 2.7 g/m?
- 4.96** Refer to Exercise 4.95. A company has a fleet of 150 Polluters used by its sales staff. Describe the distribution of the total amount, in g/m, of nitrogen oxide produced in the exhaust of this fleet. What are the mean and standard deviation of the total amount, in g/m, of nitrogen oxide in the exhaust for the fleet? (*Hint:* The total amount of nitrogen oxide can be represented as $\sum_{i=1}^{150} W_i$, where W_i is the amount of nitrogen oxide in the exhaust of the i th car. Thus, the Central Limit Theorem for sums is applicable.)

- Soc. 4.97** The baggage limit for an airplane is set at 100 pounds per passenger. Thus, for an airplane with 200 passenger seats there would be a limit of 20,000 pounds. The weight of the baggage of an individual passenger is a random variable with a mean of 95 pounds and a standard deviation of 35 pounds. If all 200 seats are sold for a particular flight, what is the probability that the total weight of the passengers' baggage will exceed the 20,000-pound limit?
- Med. 4.98** A patient visits her doctor with concerns about her blood pressure. If the systolic blood pressure exceeds 150, the patient is considered to have high blood pressure and medication may be prescribed. The problem is that there is a considerable variation in a patient's systolic blood pressure readings during a given day.
- If a patient's systolic readings during a given day have a normal distribution with a mean of 160 mm mercury and a standard deviation of 20 mm, what is the probability that a single measurement will fail to detect that the patient has high blood pressure?
 - If five measurements are taken at various times during the day, what is the probability that the average blood pressure reading will be less than 150 and hence fail to indicate that the patient has a high blood pressure problem?
 - How many measurements would be required so that the probability is at most 1% of failing to detect that the patient has high blood pressure?

4.13 Normal Approximation to the Binomial

A binomial random variable y was defined earlier to be the number of successes observed in n independent trials of a random experiment in which each trial resulted in either a success (S) or a failure (F) and $P(S) = \pi$ for all n trials. We will now demonstrate how the Central Limit Theorem for sums enables us to calculate probabilities for a binomial random variable by using an appropriate normal curve as an approximation to the binomial distribution. We said in Section 4.8 that probabilities associated with values of y can be computed for a binomial experiment for any values of n or π , but the task becomes more difficult when n gets large. For example, suppose a sample of 1,000 voters is polled to determine sentiment toward the consolidation of city and county government. What would be the probability of observing 460 or fewer favoring consolidation if we assume that 50% of the entire population favor the change? Here we have a binomial experiment with $n = 1,000$ and π , the probability of selecting a person favoring consolidation, equal to .5. To determine the probability of observing 460 or fewer favoring consolidation in the random sample of 1,000 voters, we could compute $P(y)$ using the binomial formula for $y = 460, 459, \dots, 0$. The desired probability would then be

$$P(y = 460) + P(y = 459) + \dots + P(y = 0)$$

There would be 461 probabilities to calculate with each one being somewhat difficult because of the factorials. For example, the probability of observing 460 favoring consolidation is

$$P(y = 460) = \frac{1000!}{460!540!} (.5)^{460} (.5)^{540}$$

A similar calculation would be needed for all other values of y .

To justify the use of the Central Limit Theorem, we need to define n random variables, $I_1, \dots, I_n$, by

$$I_i = \begin{cases} 1 & \text{if the } i\text{th trial results in a success} \\ 0 & \text{if the } i\text{th trial results in a failure} \end{cases}$$

The binomial random variable y is the number of successes in the n trials. Now, consider the sum of the random variable $I_1, \dots, I_n, \sum_{i=1}^n I_i$. A 1 is placed in the sum for each S that occurs and a 0 for each F that occurs. Thus, $\sum_{i=1}^n I_i$ is the number of S's that occurred during the n trials. Hence, we conclude that $y = \sum_{i=1}^n I_i$. Because the binomial random variable y is the sum of independent random variables, each having the same distribution, we can apply the Central Limit Theorem for sums to y . Thus, the normal distribution can be used to approximate the binomial distribution when n is of an appropriate size. The normal distribution that will be used has a mean and standard deviation given by the following formula:

$$\mu = n\pi \quad \sigma = \sqrt{n\pi(1-\pi)}$$

These are the mean and standard deviation of the binomial random variable y .

EXAMPLE 4.21

Use the normal approximation to the binomial to compute the probability of observing 460 or fewer in a sample of 1,000 favoring consolidation if we assume that 50% of the entire population favor the change.

Solution The normal distribution used to approximate the binomial distribution will have

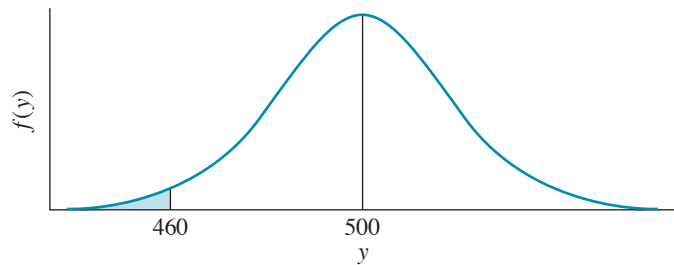
$$\mu = n\pi = 1,000(.5) = 500$$

$$\sigma = \sqrt{n\pi(1-\pi)} = \sqrt{1,000(.5)(.5)} = 15.8$$

The desired probability is represented by the shaded area shown in Figure 4.25. We calculate the desired area by first computing

$$z = \frac{y - \mu}{\sigma} = \frac{460 - 500}{15.8} = -2.53$$

FIGURE 4.25
Approximating normal
distribution for the binomial
distribution, $\mu = 500$ and
 $\sigma = 15.8$



Referring to Table 1 in the Appendix, we find that the area under the normal curve to the left of 460 (for $z = -2.53$) is .0057. Thus, the probability of observing 460 or fewer favoring consolidation is approximately .0057.

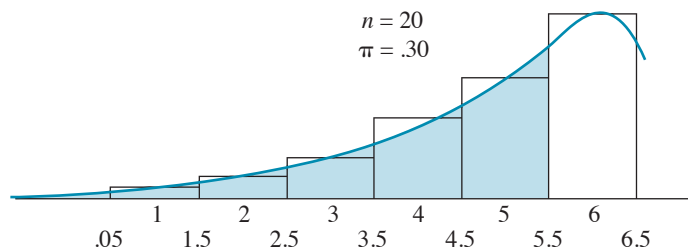
The normal approximation to the binomial distribution can be unsatisfactory if $n\pi < 5$ or $n(1 - \pi) < 5$. If π , the probability of success, is small, and n , the sample size, is modest, the actual binomial distribution is seriously skewed to the right. In such a case, the symmetric normal curve will give an unsatisfactory approximation. If π is near 1, so $n(1 - \pi) < 5$, the actual binomial will be skewed to the left, and again the normal approximation will not be very accurate. The normal approximation, as described, is quite good when $n\pi$ and $n(1 - \pi)$ exceed about 20. In the middle zone, $n\pi$ or $n(1 - \pi)$ between 5 and 20, a modification

continuity correction

called a **continuity correction** makes a substantial contribution to the quality of the approximation.

The point of the continuity correction is that we are using the continuous normal curve to approximate a discrete binomial distribution. A picture of the situation is shown in Figure 4.26.

FIGURE 4.26
Normal approximation
to binomial



The binomial probability that $y \leq 5$ is the sum of the areas of the rectangle above 5, 4, 3, 2, 1, and 0. This probability (area) is approximated by the area under the superimposed normal curve to the left of 5. Thus, the normal approximation ignores half of the rectangle above 5. The continuity correction simply includes the area between $y = 5$ and $y = 5.5$. For the binomial distribution with $n = 20$ and $\pi = .30$ (pictured in Figure 4.26), the correction is to take $P(y \leq 5)$ as $P(y \leq 5.5)$. Instead of

$$P(y \leq 5) = P[z \leq (5 - 20(.3))/\sqrt{20(.3)(.7)}] = P(z \leq -.49) = .3121$$

use

$$P(y \leq 5.5) = P[z \leq (5.5 - 20(.3))/\sqrt{20(.3)(.7)}] = P(z \leq -.24) = .4052$$

The actual binomial probability can be shown to be .4164. The general idea of the continuity correction is to add or subtract .5 from a binomial value before using normal probabilities. The best way to determine whether to add or subtract is to draw a picture like Figure 4.26.

**Normal Approximation to
the Binomial Probability
Distribution**

For large n and π not too near 0 or 1, the distribution of a binomial random variable y may be approximated by a normal distribution with $\mu = n\pi$ and $\sigma = \sqrt{n\pi(1-\pi)}$. This approximation should be used only if $n\pi \geq 5$ and $n(1-\pi) \geq 5$. A continuity correction will improve the quality of the approximation in cases in which n is not overwhelmingly large.

EXAMPLE 4.22

A large drug company has 100 potential new prescription drugs under clinical test. About 20% of all drugs that reach this stage are eventually licensed for sale. What is the probability that at least 15 of the 100 drugs are eventually licensed? Assume that the binomial assumptions are satisfied, and use a normal approximation with continuity correction.

Solution The mean of y is $\mu = 100(.2) = 20$; the standard deviation is $\sigma = \sqrt{100(.2)(.8)} = 4.0$. The desired probability is that 15 or more drugs are approved.

Because $y = 15$ is included, the continuity correction is to take the event as y greater than or equal to 14.5.

$$\begin{aligned} P(y \geq 14.5) &= P\left(z \geq \frac{14.5 - 20}{4.0}\right) = P(z \geq -1.38) = 1 - P(z < -1.38) \\ &= 1 - .0838 = .9162 \end{aligned}$$

4.14 Minitab Instructions

Generating Random Numbers

To generate 1,000 random numbers from the set $[0, 1, \dots, 9]$,

1. Click on **Calc**, then **Random Data**, then **Integer**.
2. Type the number of rows of data: **Generate 20 rows of data**.
3. Type the columns in which the data are to be stored: **Store in column(s): c1—c50**.
4. Type the first number in the list: **Minimum value: 0**.
5. Type the last number in the list: **Maximum value: 9**.
6. Click on **OK**.

Note that we have generated $(20)(50) = 1,000$ random numbers.

Calculating Binomial Probabilities

To calculate binomial probabilities when $n = 10$ and $\pi = 0.6$,

1. Enter the values of x in column c1: **0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10**.
2. Click on **Calc**, then **Probability Distributions**, then **Binomial**.
3. Select either **Probability** [to compute $P(X = x)$] or **Cumulative probability** [to compute $P(X \leq x)$].
4. Type the value of n : **Number of trials: 10**.
5. Type the value of π : **Probability of success: 0.6**.
6. Click on **Input column**.
7. Type the column number where values of x are located: **C1**.
8. Click on **Optional storage**.
9. Type the column number to store probability: **C2**.
10. Click on **OK**.

Calculating Normal Probabilities

To calculate $P(X \leq 18)$ when X is normally distributed with $\mu = 23$ and $\sigma = 5$,

1. Click on **Calc**, then **Probability Distributions**, then **Normal**.
2. Click on **Cumulative probability**.
3. Type the value of μ : **Mean: 23**.
4. Type the value of σ : **Standard deviation: 5**.
5. Click on **Input constant**.
6. Type the value of x : **18**.
7. Click on **OK**.

Generating Sampling Distribution of $\bar{y}$

To create the sampling distribution of $\bar{y}$ based on 500 samples of size $n = 16$ from a normal distribution with $\mu = 60$ and $\sigma = 5$,

1. Click on **Calc**, then **Random Data**, then **Normal**.
2. Type the number of samples: **Generate 500 rows**.
3. Type the sample size n in terms of number of columns: **Store in column(s) c1-c16**.
4. Type in the value of μ : **Mean: 60**.
5. Type in the value of σ : **Standard deviation: 5**.
6. Click on **OK**. There are now 500 rows in columns c1–c16, 500 samples of 16 values each to generate 500 values of $\bar{y}$.
7. Click on **Calc**, then **Row Statistics**, then **mean**.
8. Type in the location of data: **Input Variables c1-c16**.
9. Type in the column in which the 500 means will be stored: **Store Results in c17**.
10. To obtain the mean of the 500 $\bar{y}$ s, click on **Calc**, then **Column Statistics**, then **mean**.
11. Type in the location of the 500 means: **Input Variables c17**.
12. Click on **OK**.
13. To obtain the standard deviation of the 500 $\bar{y}$ s, click on **Calc**, then **Column Statistics**, then **standard deviation**.
14. Type in the location of the 500 means: **Input Variables c17**.
15. Click on **OK**.
16. To obtain the sampling distribution of $\bar{y}$, click **Graph**, then **Histogram**.
17. Type **c17** in the Graph box.
18. Click on **OK**.

4.15 Summary

In this chapter, we presented an introduction to probability, probability distributions, and sampling distributions. Knowledge of the probabilities of sample outcomes is vital to a statistical inference. Three different interpretations of the probability of an outcome were given: the classical, relative frequency, and subjective interpretations. Although each has a place in statistics, the relative frequency approach has the most intuitive appeal because it can be checked.

Quantitative random variables are classified as either discrete or continuous random variables. The probability distribution for a discrete random variable y is a display of the probability $P(y)$ associated with each value of y . This display may be presented in the form of a histogram, table, or formula.

The binomial is a very important and useful discrete random variable. Many experiments that scientists conduct are similar to a coin-tossing experiment where dichotomous (yes–no) type data are accumulated. The binomial experiment frequently provides an excellent model for computing probabilities of various sample outcomes.

Probabilities associated with a continuous random variable correspond to areas under the probability distribution. Computations of such probabilities were illustrated for areas under the normal curve. The importance of this exercise is

borne out by the Central Limit Theorem: Any random variable that is expressed as a sum or average of a random sample from a population having a finite standard deviation will have a normal distribution for a sufficiently large sample size. Direct application of the Central Limit Theorem gives the sampling distribution for the sample mean. Because many sample statistics are either sums or averages of random variables, application of the Central Limit Theorem provides us with information about probabilities of sample outcomes. These probabilities are vital for the statistical inferences we wish to make.

Key Formulas

1. Binomial probability distribution

$$P(y) = \frac{n!}{y!(n-y)!} \pi^y (1-\pi)^{n-y}$$

2. Sampling distribution for $\bar{y}$

Mean: μ

Standard error: $\sigma_{\bar{y}} = \sigma/\sqrt{n}$

3. Normal approximation to the binomial

$$\mu = n\pi \quad \sigma = \sqrt{n\pi(1-\pi)}$$

provided that $n\pi$ and $n(1-\pi)$ are greater than or equal to 5 or, equivalently, if

$$n \geq \frac{5}{\min(\pi, 1-\pi)}$$

Supplementary Exercises

- Bus. 4.99** One way to audit expense accounts for a large consulting firm is to sample all reports dated the last day of each month. Comment on whether such a sample constitutes a random sample.
- Bus. 4.100** Critical key-entry errors in the data processing operation of a large district bank occur approximately .1% of the time. If a random sample of 10,000 entries is examined, determine the following:
- the expected number of errors
 - the probability of observing fewer than five errors
 - the probability of observing fewer than two errors
- 4.101** Use the binomial distribution with $n = 20$, $\pi = .5$ to compare accuracy of the normal approximation to the binomial.
- Compute the exact probabilities and corresponding normal approximations for $y < 5$.
 - The normal approximation can be improved slightly by taking $P(y \leq 4.5)$. Why should this help? Compare your results.
 - Compute the exact probabilities and corresponding normal approximations with the continuity correction for $P(8 < y < 14)$.
- 4.102** Let y be a binomial random variable with $n = 10$ and $\pi = .5$.
- Calculate $P(4 \leq y \leq 6)$.
 - Use a normal approximation without the continuity correction to calculate the same probability. Compare your results. How well did the normal approximation work?
- 4.103** Refer to Exercise 4.102. Use the continuity correction to compute the probability $P(4 \leq y \leq 6)$. Does the continuity correction help?
- Bus. 4.104** A marketing research firm believes that approximately 25% of all persons mailed a sweepstakes offer will respond if a preliminary mailing of 5,000 is conducted in a fixed region.
- What is the probability that 1,000 or fewer will respond?
 - What is the probability that 3,000 or more will respond?

Engin. **4.105** The breaking strengths for 1-foot-square samples of a particular synthetic fabric are approximately normally distributed with a mean of 2,250 pounds per square inch (psi) and a standard deviation of 10.2 psi.

- Find the probability of selecting a 1-foot-square sample of material at random that on testing would have a breaking strength in excess of 2,265 psi.
- Describe the sampling distribution for $\bar{y}$ based on random samples of 15 1-foot sections.

4.106 Refer to Exercise 4.105. Suppose that a new synthetic fabric has been developed that may have a different mean breaking strength. A random sample of 15 one-foot sections is obtained and each section is tested for breaking strength. If we assume that the population standard deviation for the new fabric is identical to that for the old fabric, give the standard deviation for the sampling distribution of $\bar{y}$ using the new fabric.

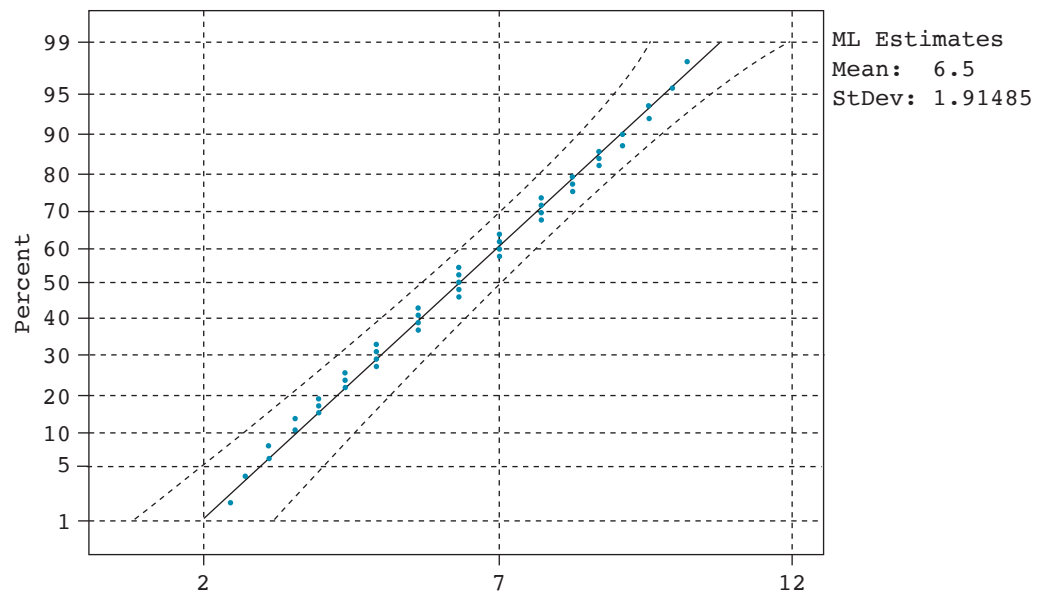
4.107 Refer to Exercise 4.106. Suppose that the mean breaking strength for the sample of 15 one-foot sections of the new synthetic fabric is 2,268 psi. What is the probability of observing a value of $\bar{y}$ equal to or greater than 2,268, assuming that the mean breaking strength for the new fabric is 2,250, the same as that for the old?

4.108 Based on your answer in Exercise 4.107, do you believe the new fabric has the same mean breaking strength as the old? (Assume $\sigma = 10.2$.)

4.109 In Figure 4.19, we visually inspected the relative frequency histogram for sample means based on two measurements and noted its bell shape. Another way to determine whether a set of measurements is bell-shaped (normal) is to construct a **normal probability plot** of the sample data. If the plotted points are nearly a straight line, we say the measurements were selected from a normal population. We can generate a normal probability plot using the following Minitab code. If the plotted points fall within the curved dotted lines, we consider the data to be a random sample from a normal distribution.

Minitab code:

- Enter the 45 measurements into C1 of the data spreadsheet.
- Click on **Graph**, then **Probability Plot**.
- Type **c1** in the box labeled **Variables**.
- Click on **OK**.



The 45 data values clearly fall within the two curved lines and fairly close to the straight line. Thus, we conclude there is a strong indication that the data values follow a normal distribution.

- a. Suppose our population was the ten measurements (2, 3, 6, 8, 9, 12, 25, 29, 39, 50). Generate the 45 sample means based on $n = 2$ observations per sample and determine whether the sampling distribution of the sample mean is approximately normally distributed by constructing a histogram and normal probability plot of the 45 sample means.
- b. Why do you think the plots for the means from the population (2, 3, 6, 8, 9, 12, 25, 29, 39, 50) differ greatly from the plots obtained for the means from the population (2, 3, 4, 5, 6, 7, 8, 9, 10, 11)?

H.R. 4.110 A labor union's examining board for the selection of apprentices has a record of admitting 70% of all applicants who satisfy a set of basic requirements. Five members of a minority group recently came before the board, and four of five were rejected. Find the probability that one or fewer would be accepted if the record is really .7. Did the board apply a lower probability of acceptance when reviewing the five members of the minority group?

Gov. 4.111 Suppose that you are a regional director of the IRS office and that you are charged with sampling 1% of the returns with gross income levels above \$15,000. How might you go about this? Would you use random sampling? How?

Med. 4.112 Experts consider high serum cholesterol levels to be associated with an increased incidence of coronary heart disease. Suppose that the logarithm of cholesterol levels for males in a given age bracket are normally distributed with a mean of 2.35 and a standard deviation of .12.

- a. What percentage of the males in this age bracket could be expected to have a serum cholesterol level greater than 250 mg/ml, the upper limit of the clinical normal range?
- b. What percentage of the males could be expected to have serum cholesterol levels within the clinical normal range of 150–250 mg/ml?
- c. If levels above 300 mg/ml are considered very risky, what percentage of the adult males in this age bracket could be expected to exceed 300?

Bus. 4.113 One of the major soft-drink companies changed the secret formula for its leading beverage to attract new customers. Recently, a marketing research firm interviewed 1,000 potential new customers and, after giving them a taste of the newly reformulated beverage, determined the number of these individuals planning to buy the reformulated beverage in the near future.

- a. Identify the random variable for the population of $y =$ values of interest.
- b. Can you compute the mean and variance? Why or why not?
- c. How would you calculate $P(y \leq 250)$?

Bus. 4.114 Many firms are using or exploring the possibility of using telemarketing techniques—that is, marketing their products via the telephone to supplement the more traditional marketing strategies. Assume a firm finds that approximately 1 in every 100 calls yields a sale.

- a. Find the probability the first sale will occur somewhere in the first 5 calls.
- b. Find the probability the first sale will occur sometime after 10 calls.

Bus. 4.115 Marketing analysts have determined that a particular advertising campaign should make at least 20% of the adult population aware of the advertised product. After a recent campaign, 25 of 400 adults sampled indicated that they had seen the ad and were aware of the new product.

- a. Find the approximate probability of observing $y \leq 25$ given that 20% of the population is aware of the product through the campaign.
- b. Based on your answer to part (a), does it appear the ad was successful? Explain.

Med. 4.116 One or more specific, minor birth defects occurs with probability .0001 (that is, 1 in 10,000 births). If 20,000 babies are born in a given geographic area in a given year, can we calculate the probability of observing at least one of the minor defects using the binomial or normal approximation to the binomial? Explain.

4.117 The sample mean to be calculated from a random sample of size $n = 4$ from a population consists of the eight measurements (2, 6, 9, 12, 25, 29, 39, 50). Find the sampling distribution of $\bar{y}$. (Hint: There are 70 samples of size 4 when sampling from a population of eight measurements.)

- 4.118** Plot the sampling distribution of $\bar{y}$ from Exercise 4.115.
- Does the sampling distribution appear to be approximately normal?
 - Verify that the mean of the sampling distribution of $\bar{y}$ equals the mean of the eight population values.
- 4.119** Refer to Exercise 4.117. Use the same population to find the sampling distribution for the sample median based on samples of size $n = 4$.
- 4.120** Plot the sampling distribution of the sample median of Exercise 4.119.
- Does the sampling distribution appear to be approximately normal?
 - Compute the mean of the sampling distribution of the sample median and compare this value to the population median.
- 4.121** Random samples of size 5, 20, and 80 are drawn from a population with mean $\mu = 100$ and standard deviation $\sigma = 15$.
- Give the mean of the sampling distribution of $\bar{y}$ for each of the sample sizes 5, 20, and 80.
 - Give the standard deviation of the sampling distribution of $\bar{y}$ for each of the sample sizes 5, 20, and 80.
 - Based on the results obtained in parts (a) and (b), what do you conclude about the accuracy of using the sample mean $\bar{y}$ as an estimate of population mean μ ?
- 4.122** Refer to Exercise 4.121. To evaluate how accurately the sample mean $\bar{y}$ estimates the population mean μ , we need to know the chance of obtaining a value of $\bar{y}$ that is far from μ . Suppose it is important that the sample mean $\bar{y}$ is within 5 units of the population mean μ . Find the following probabilities for each of the three sample sizes and comment on the accuracy of using $\bar{y}$ to estimate μ .
- $P(\bar{y} \geq 105)$
 - $P(\bar{y} \leq 95)$
 - $P(95 \leq \bar{y} \leq 105)$
- 4.123** A random sample of $n = 36$ measurements is selected from a population with a mean equal to 40 and a standard deviation equal to 12.
- Describe the sampling distribution of $\bar{y}$.
 - Find $P(\bar{y} > 36)$.
 - Find $P(\bar{y} < 30)$.
 - Find the value of $\bar{y}$ (say, k) such that $P(\bar{y} > k) = .05$.
- 4.124** Refer to Exercise 4.123.
- Describe the sampling distribution for the sample sum $\sum y_i$.
 - Find $P(\sum y_i > 1,440)$.
 - Find $P(\sum y_i > 1,540)$.
 - Find the value of $\sum y_i$ (say, k) such that $P[k < \sum y_i < k_2] = .95$.
- 4.125** For each of the following situations, find the expected value and standard error of $\bar{y}$ based on a random sample of size n drawn from a population with mean μ and standard deviation σ .
- $n = 25, \mu = 10, \sigma = 10$
 - $n = 100, \mu = 10, \sigma = 10$
 - $n = 25, \mu = 10, \sigma = 20$
 - $n = 100, \mu = 10, \sigma = 20$
- 4.126** Based on the results of Exercise 4.125, speculate on the effect of increasing the sample size and on the effect of an increase in σ on the standard error of $\bar{y}$.